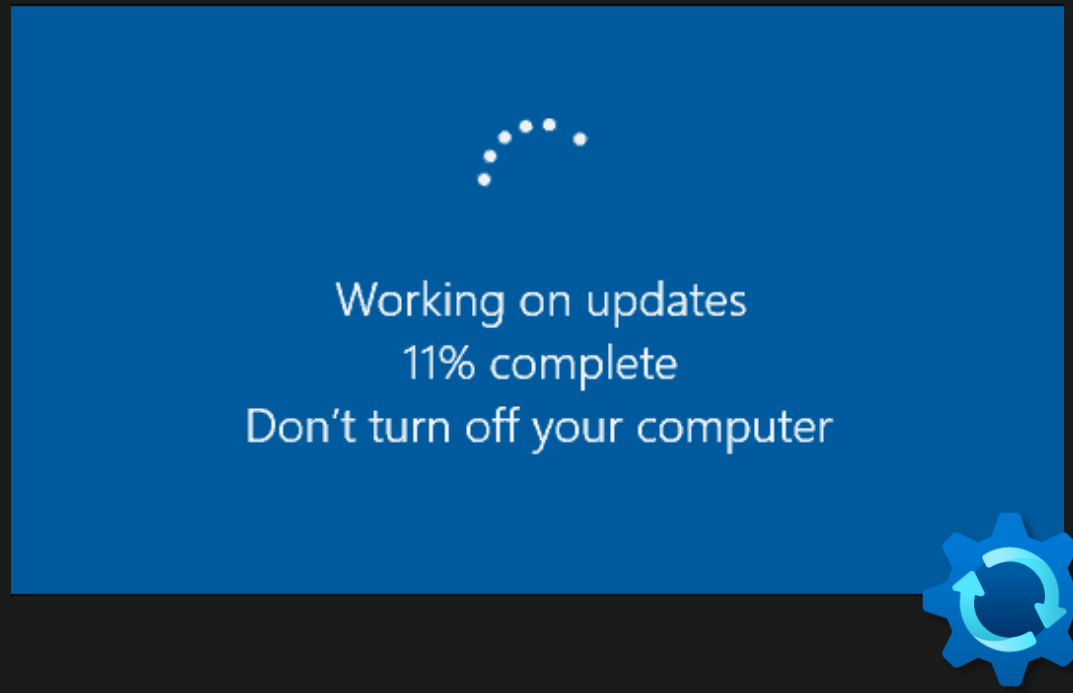
Wim Matthyssen

Behind the Patch: A Deep Dive into Azure Update Manager

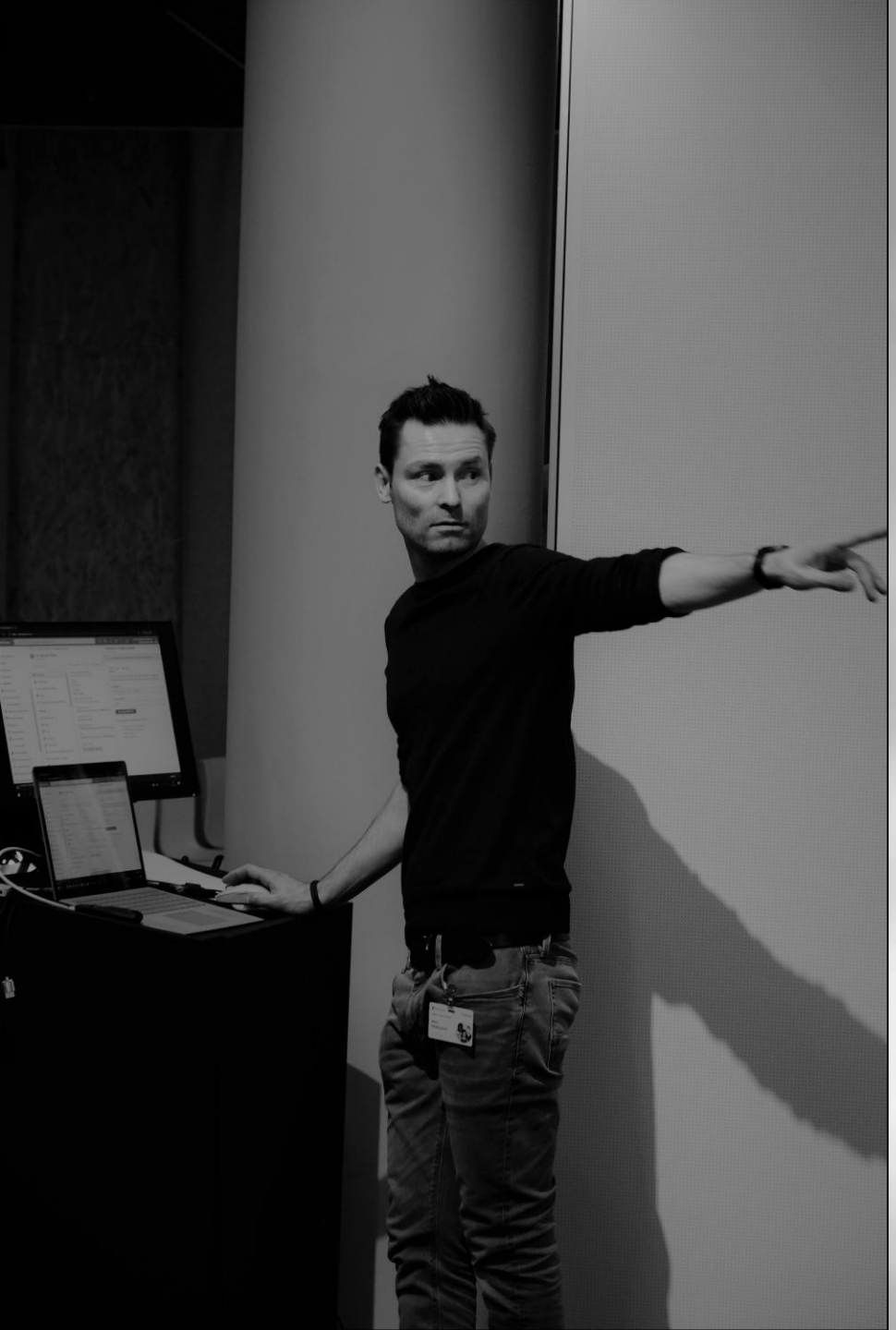
NIC26

Working on agenda updates

- How it all began …
- The Azure way
- And what about on-premises?
- Q&A



NIC26

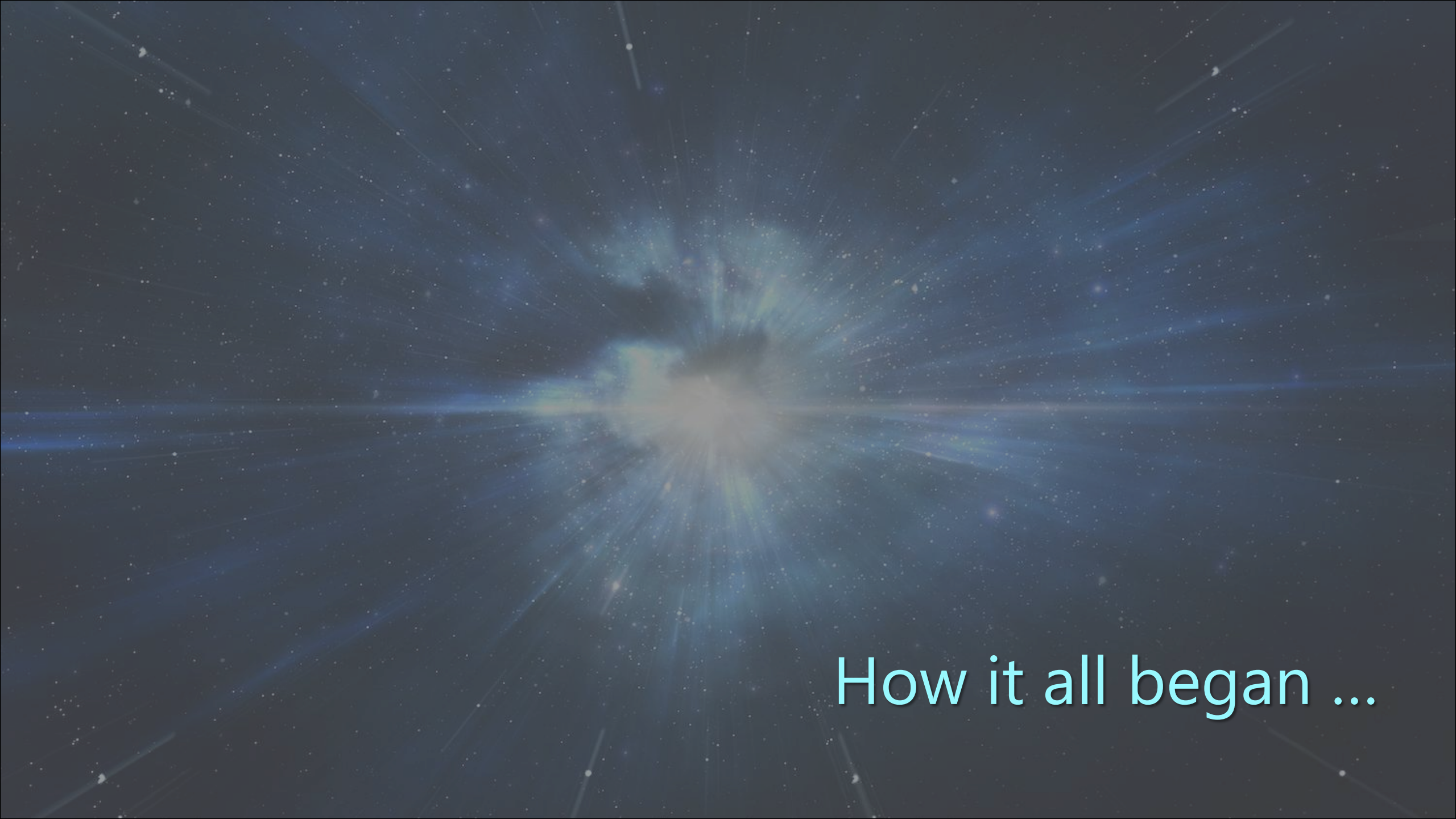


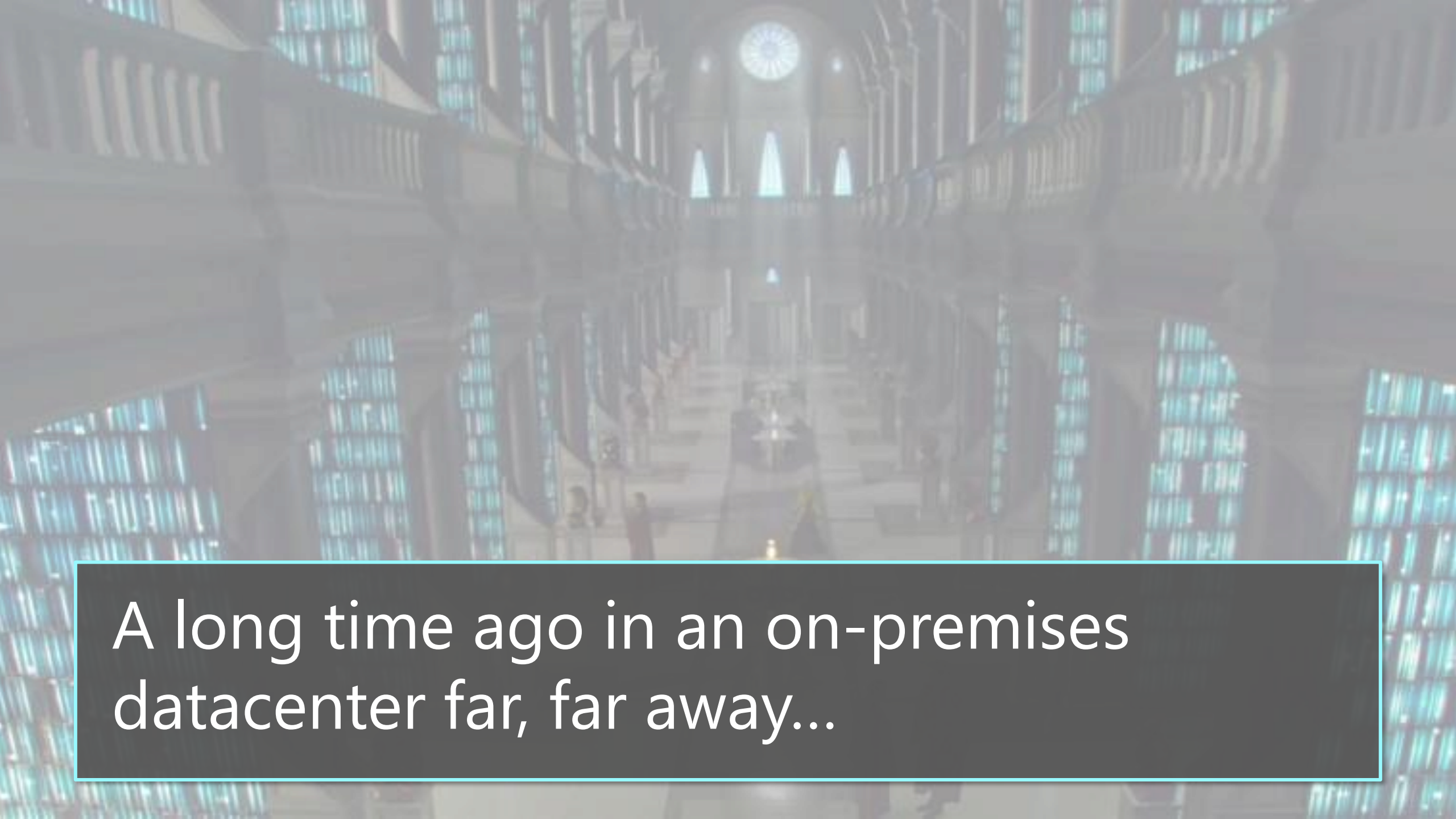
Wim Matthyssen

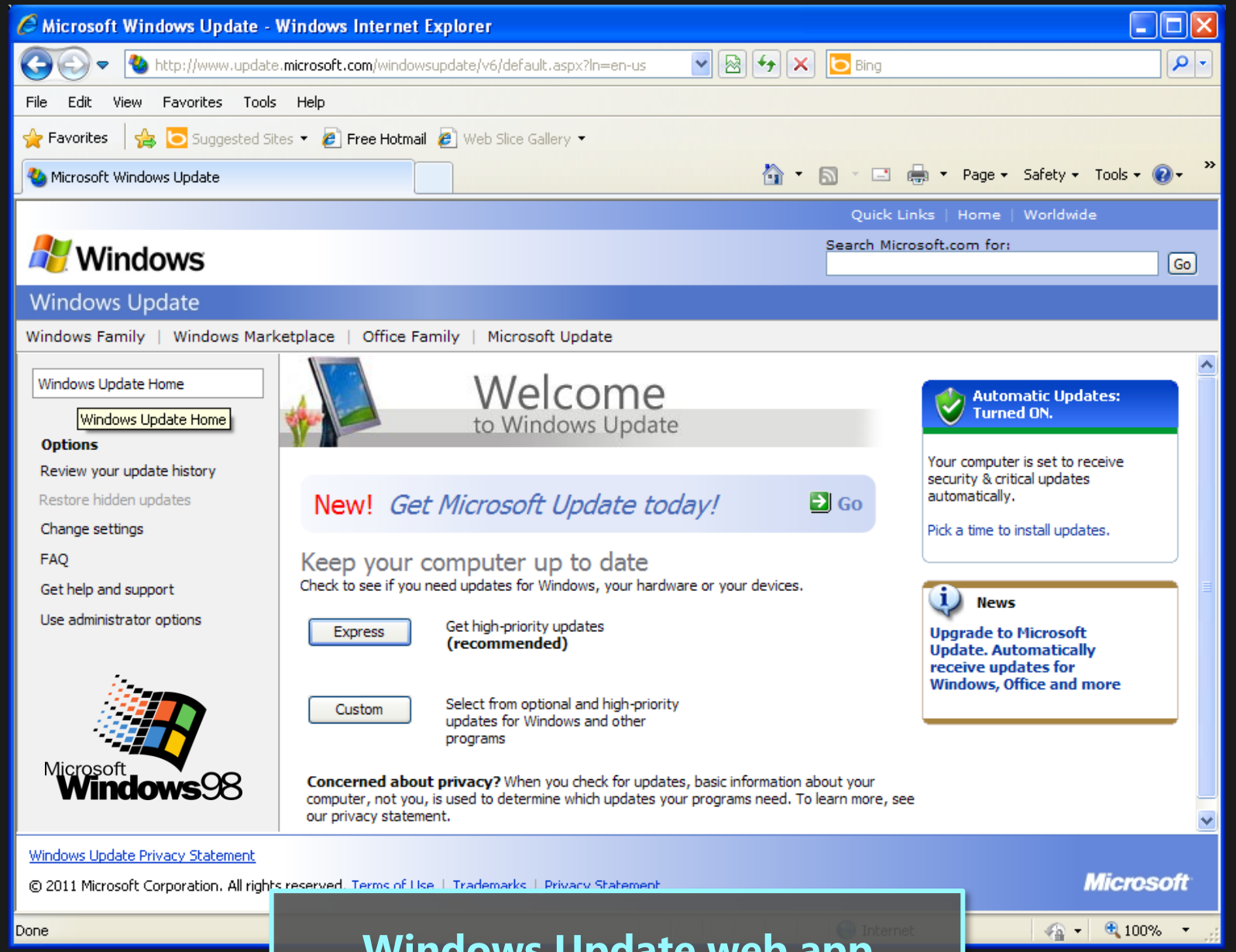
Azure Technical Advisor & Trainer @ Proximus NXT
Board member MC2MC
Microsoft Certified Trainer
Microsoft MVP Azure

@wmatthyssen
https://wmatthyssen.com
https://www.linkedin.com/in/wim-matthyssen-9053737

SCAN ME

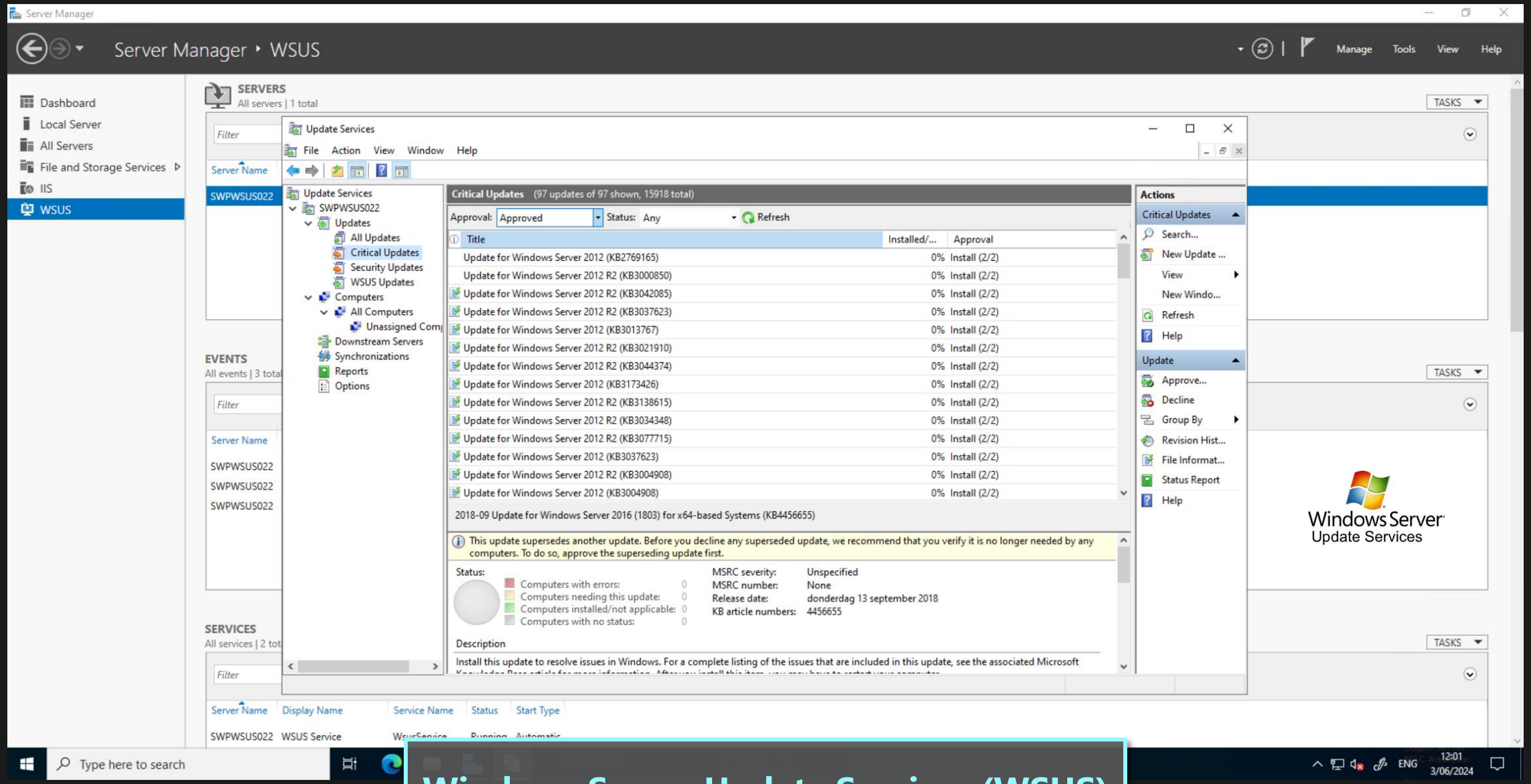

How it all began …


A long time ago in an on-premises datacenter far, far away…



Windows Update web app

NIC26



Windows Server Update Services (WSUS)

NIC26

WINDOWS IT PRO BLOG 2 MIN READ

Windows Server Update Services (WSUS) deprecation



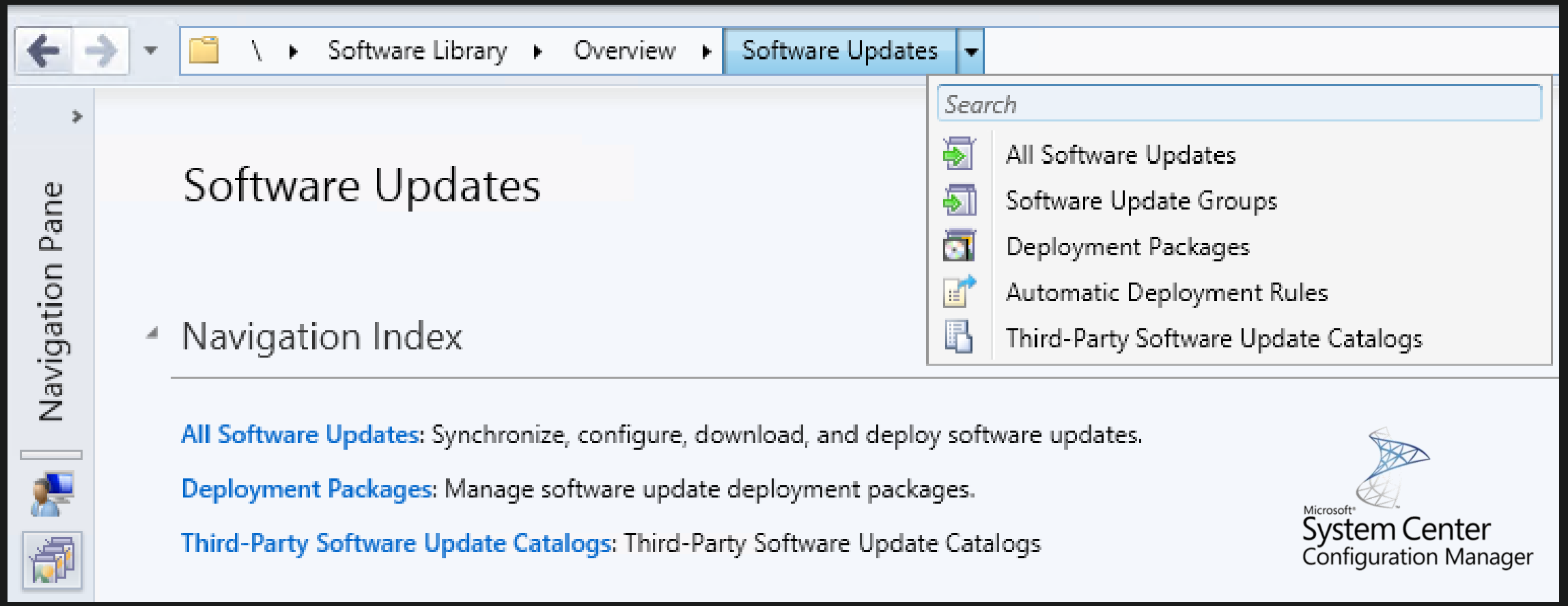
Nir_Froimovici MICROSOFT
Sep 20, 2024

Editor's note 9.25.2024: This post has been updated to define deprecation and provide clarity on how this impacts WSUS.

As part of our vision for simplified Windows management from the cloud, Microsoft has announced deprecation of Windows Server Update Services (WSUS). Specifically, this means that we are no longer investing in new capabilities, nor are we accepting new feature requests for WSUS. However, we are preserving current functionality and will continue to publish updates through the WSUS channel. We will also support any content already published through the WSUS channel.

Windows Server Update Services (WSUS) deprecation


~~A long time ago~~ In an ~~galaxy~~ hybrid ~~far, far away~~ cross
~~a long time ago~~ really not that far, far away...



Configuration Manager

NIC26

OS Patching	➡️	Azure Update Manager
Configuration	➡️	Azure Machine Configuration
Reporting	➡️	Azure Change Tracking and Inventory
Scripting	➡️	Run Command
Software Distribution	➡️	Virtual Machine Apps (VM Apps)

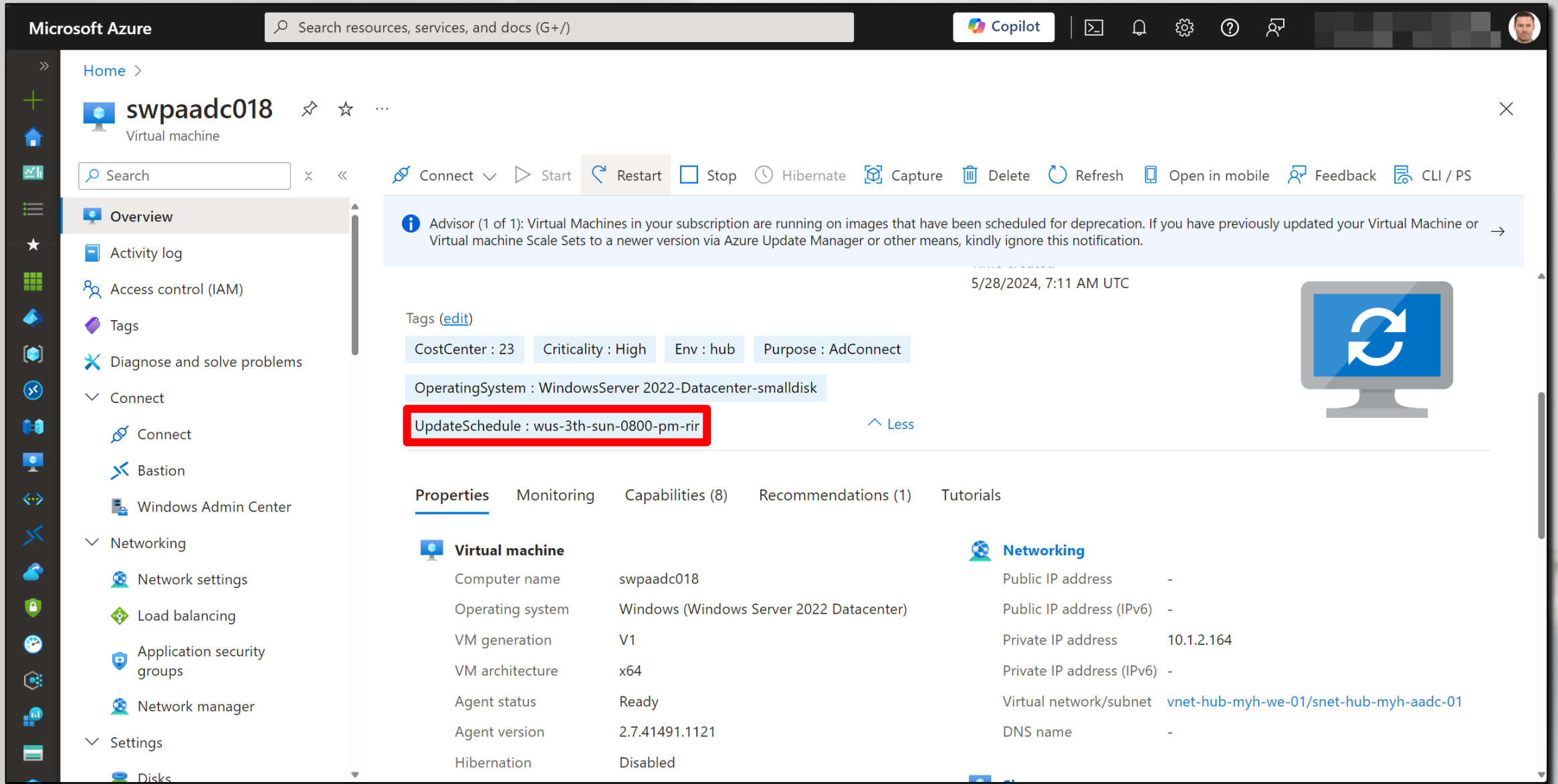
Modernize server management from Configuration Manager to Azure Arc

NIC26

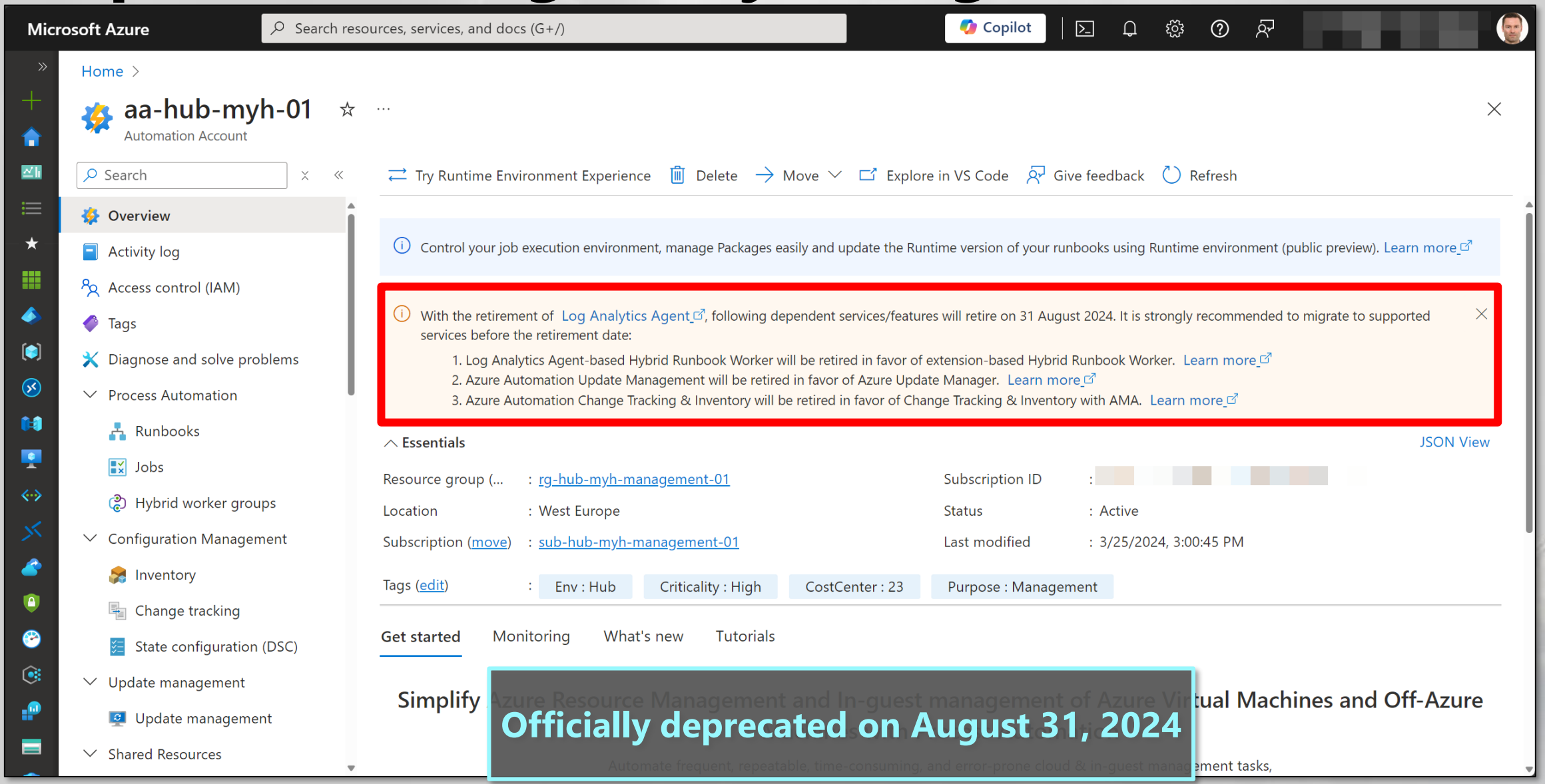
How to modernize server management from Configuration Manager to Azure Arc

The Azure Way

Azure Automation Update Management (v1)



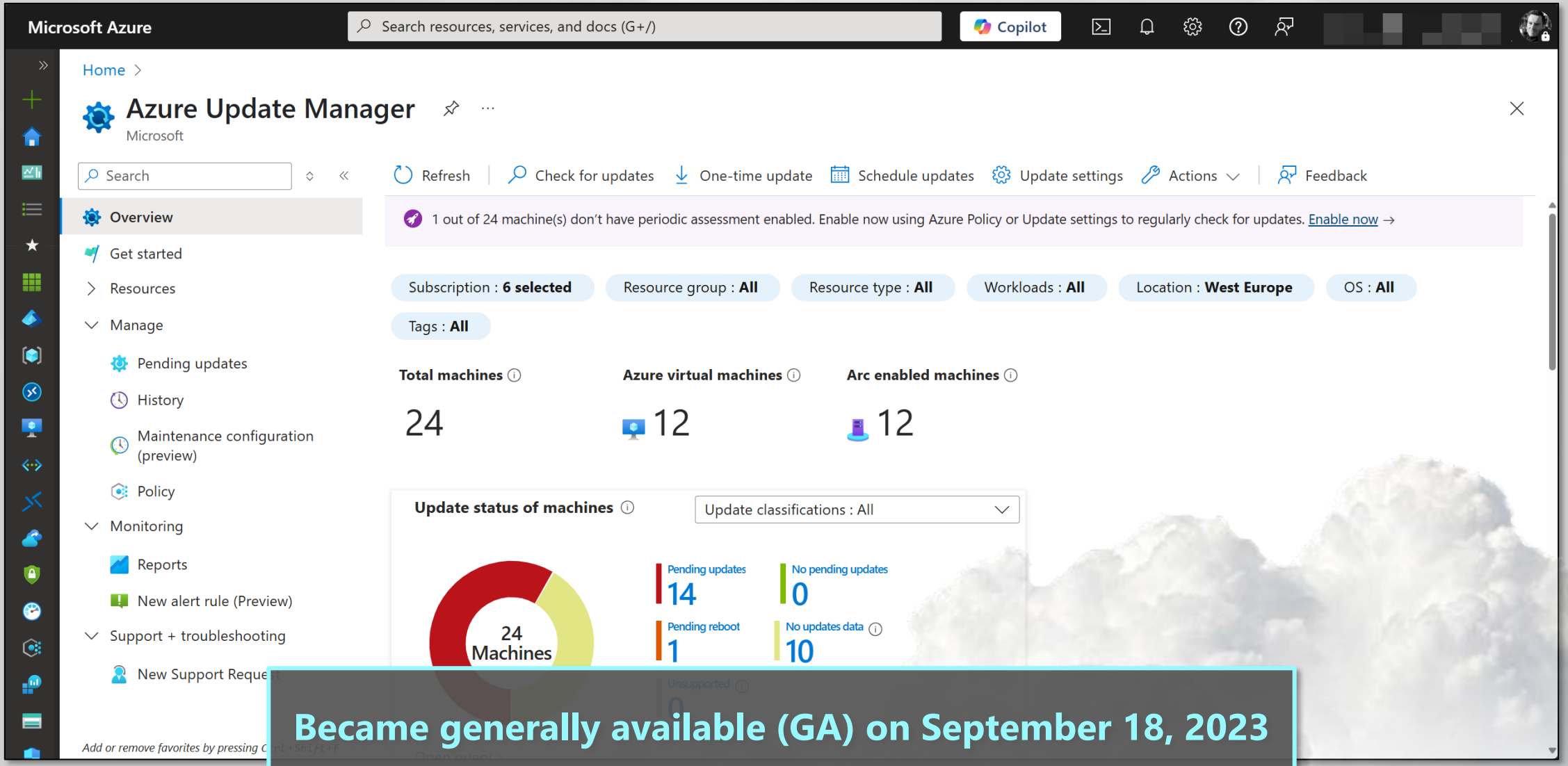
Deprecation Log Analytics agent



V1 limited capabilities

- Basic scheduling
- Manage different groups: Azure VMs and non-Azure VMs
- No on-demand capabilities
- No granular RBAC (Log Analytics or Azure Automation Account)
- Dependent on Log Analytics agent -> officially deprecated on August 31, 2024

Azure Update Manager (v2)



Became generally available (GA) on September 18, 2023

Azure Update Manager (v2)

- SaaS service
- Multi-subscription support
- Not linked to Log Analytics
- Rich scheduling
- Azure Resource Graph
- Microsoft application updates (SQL, …)
- Extension-based (Azure Arc-enabled server)
- VM-level RBAC



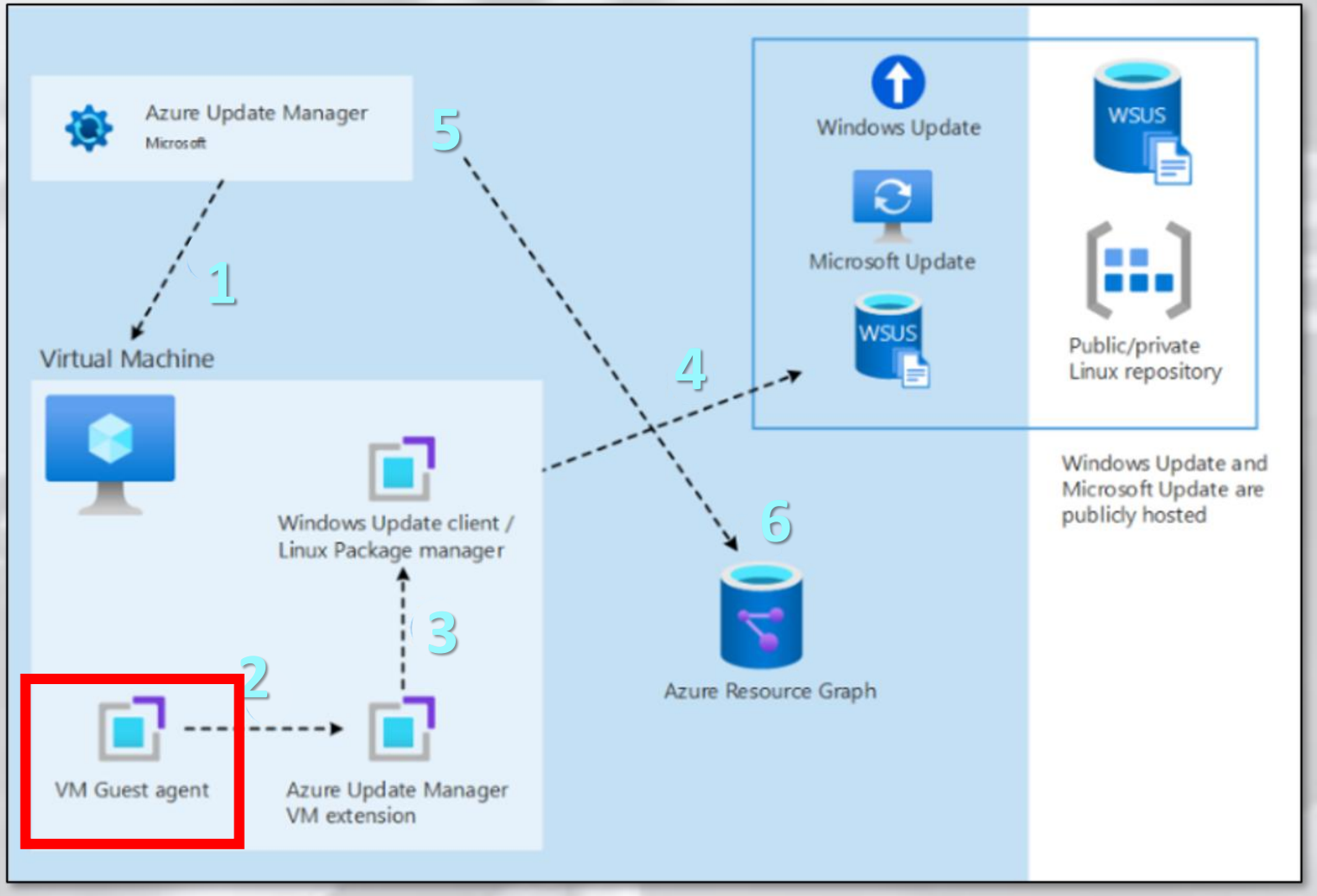
NIC26

Supports

- Windows 2008 R2 and newer Windows Server OSes (x64)
- Red Hat, Ubuntu Pro, CentOS (reaching End of Service status) and other Linux OSes
- Azure images in Azure Compute Gallery
- Azure VMs
- Azure Arc-enabled servers outside Azure
- Third party application updates (import and publish 3-party applications and custom updates to WSUS)
- Driver updates (AUM can display driver updates during assessments, but it currently cannot install them.

Does not support → x86 Windows Server OSes, Windows clients (Windows 10, 11), AVD session hosts, driver updates, Azure Kubernetes Service nodes, Virtual Machine Scale Sets (use Automatic upgrades)

How Azure Update Manager works



1. **Initiate update operation**

   Azure Update Manager targets the virtual machine.

2. **Deploy and trigger extension**

   The VM Guest Agent triggers the AUM VM extension.

3. **Invoke local update client**

   The extension calls the Windows Update client or Linux package manager.

4. **Query configured update source**

   The client checks Windows Update, Microsoft Update, WSUS, or a Linux repository.

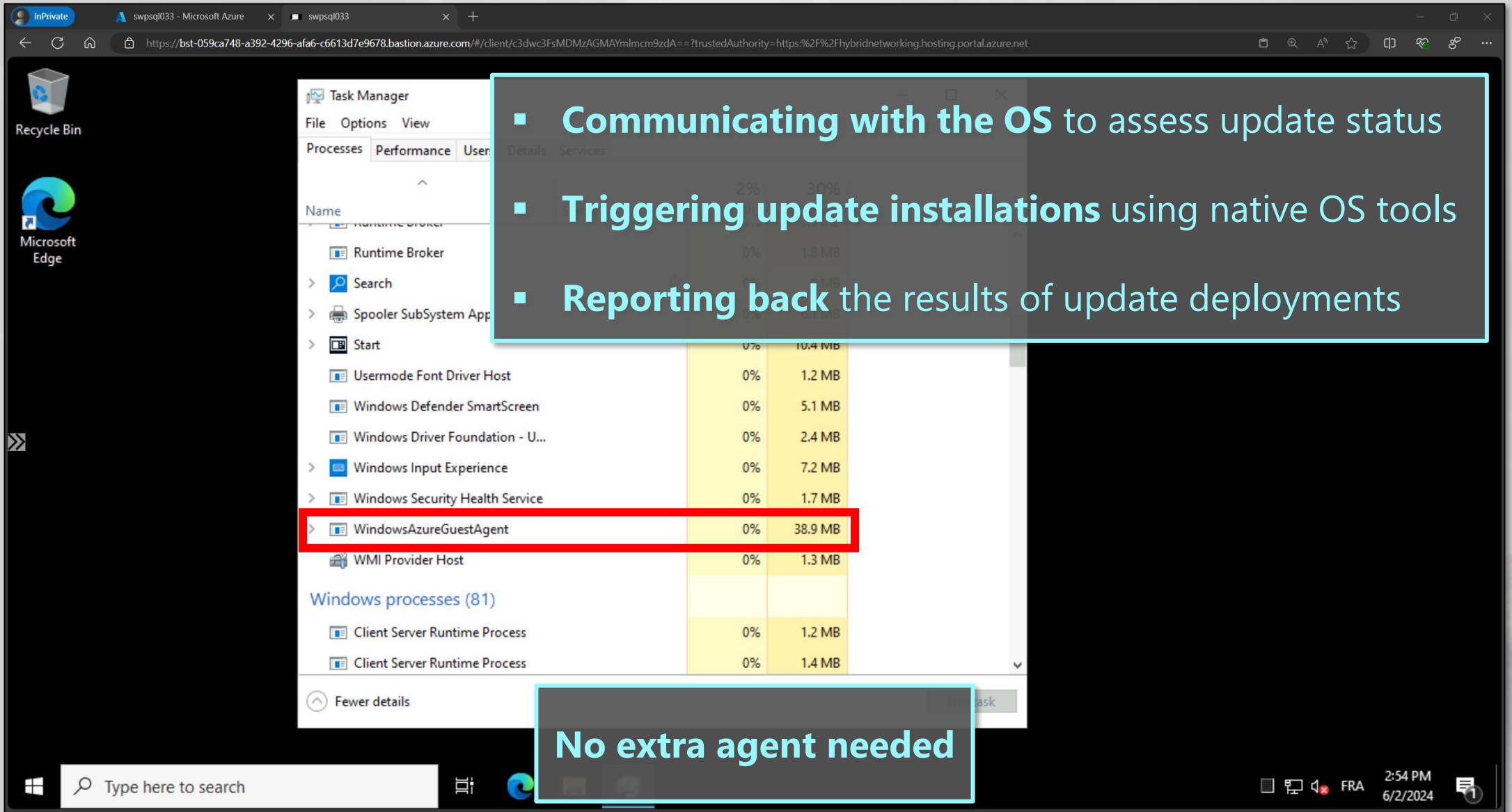
5. **Receive compliance data**

   Update Manager receives assessment and deployment compliance data.

6. **Store compliance date**

   Assessment and deployment compliance data is stored in Azure Resource Graph.

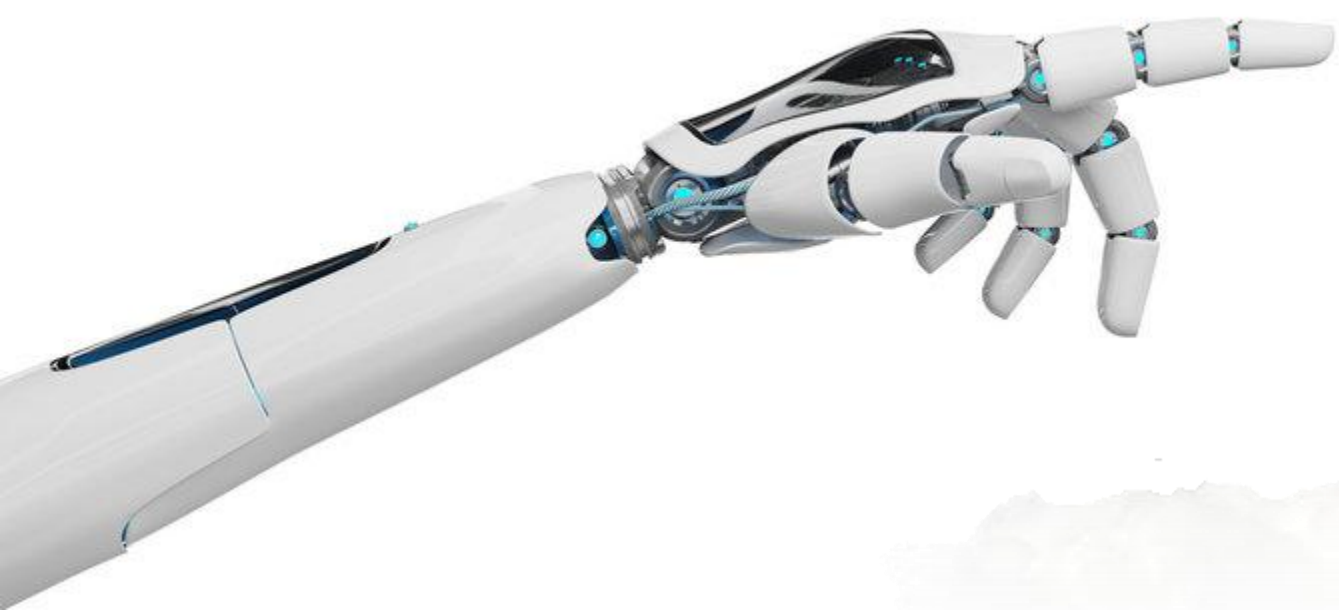
VM Agent



- **Communicating with the OS** to assess update status
- **Triggering update installations** using native OS tools
- **Reporting back** the results of update deployments

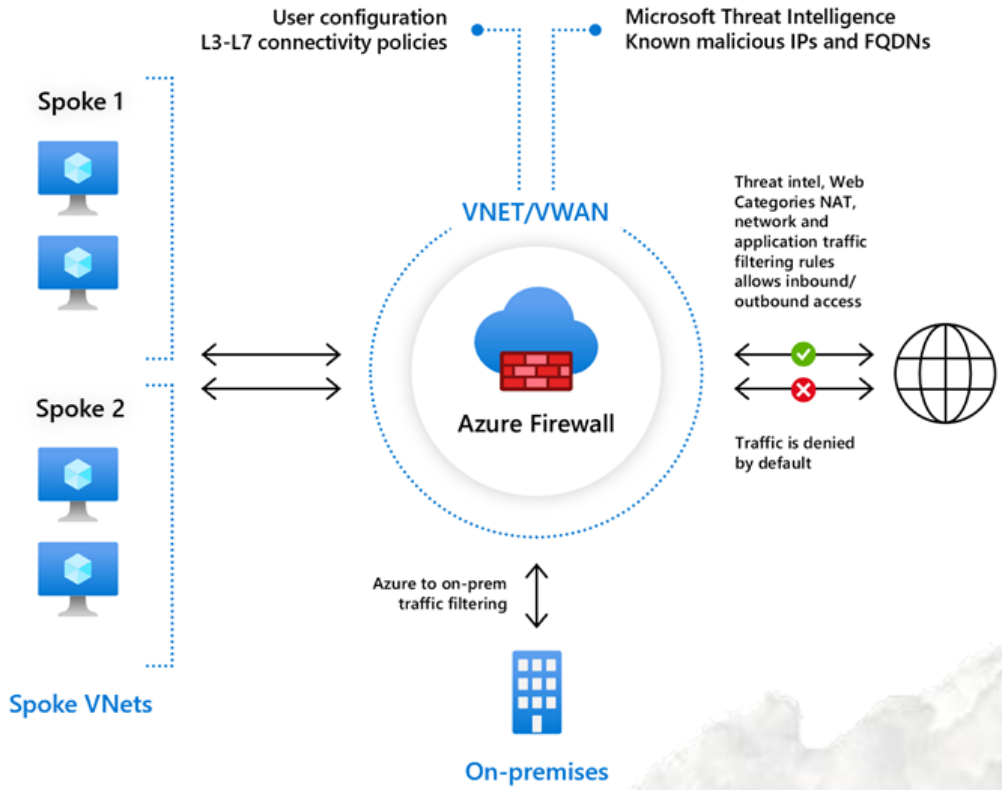
No extra agent needed

Retirement of default outbound Internet access



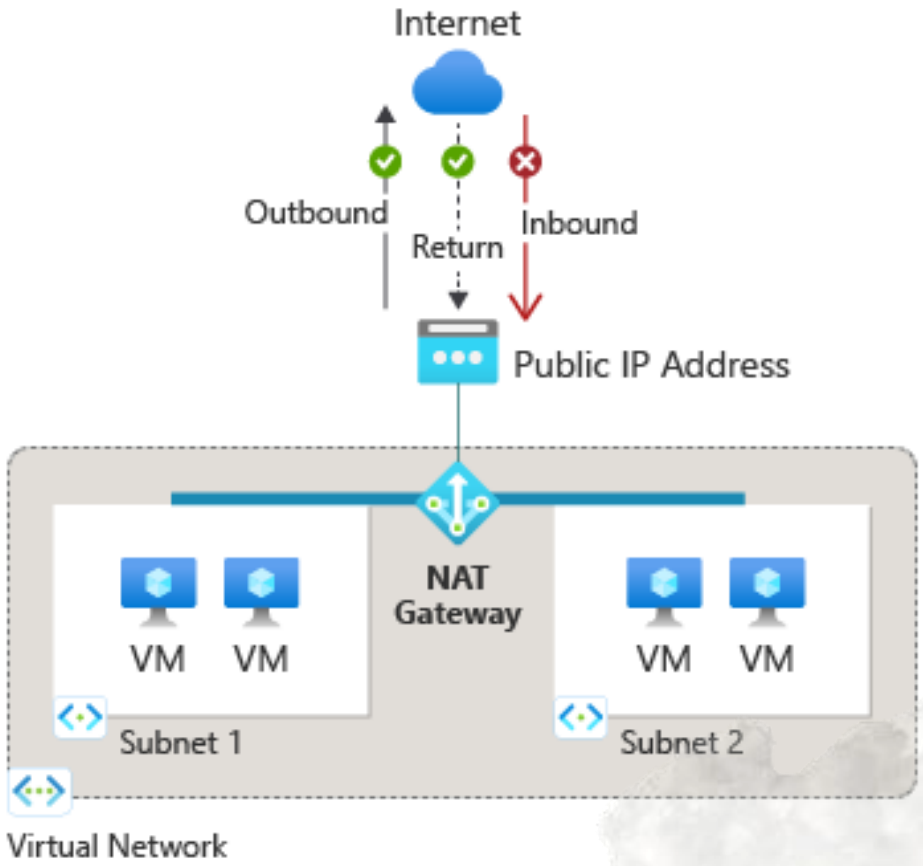
March 31, 2026

Retirement of default outbound Internet access



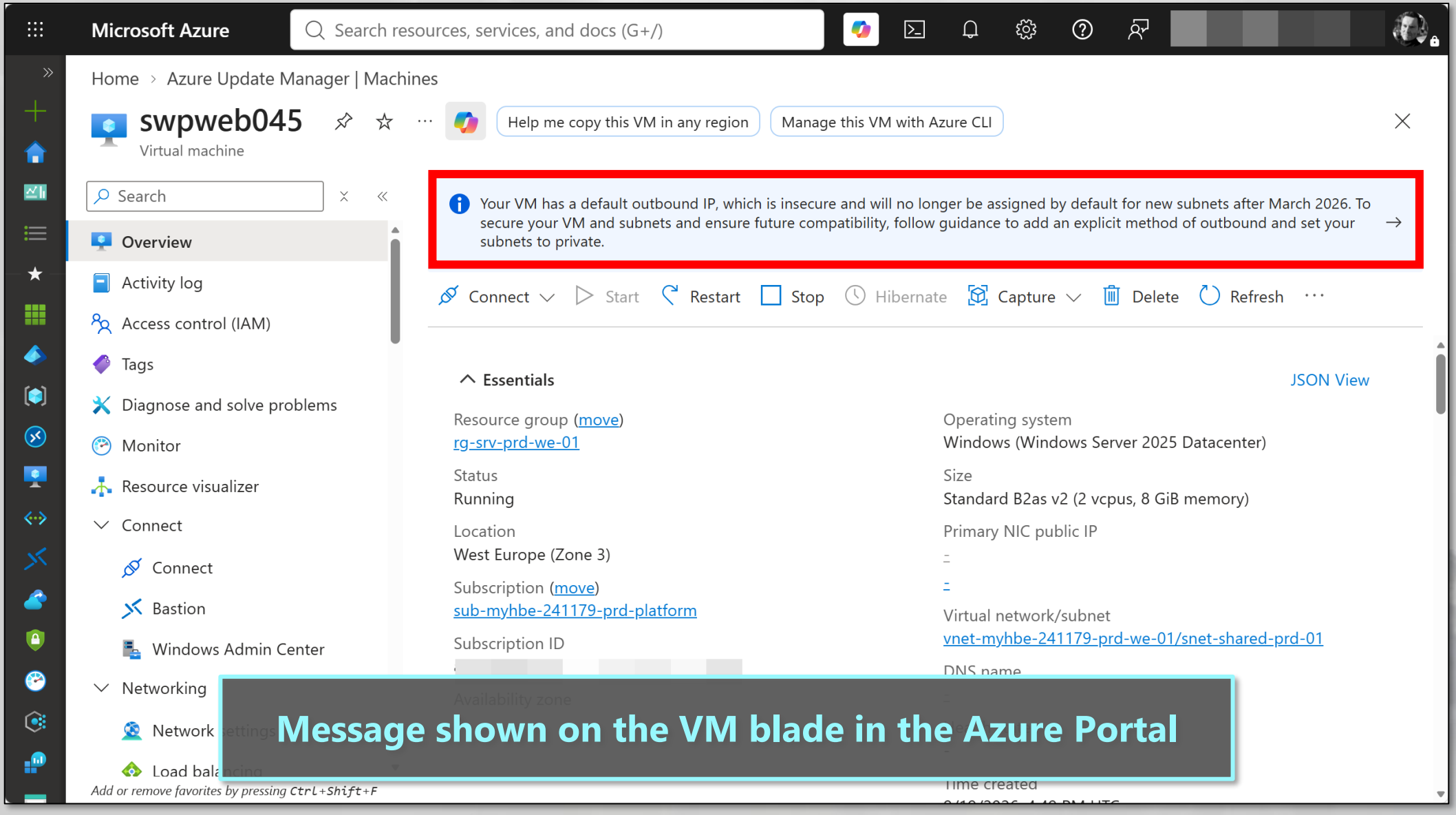
Use Azure Firewall for all egress traffic

Retirement of default outbound Internet access

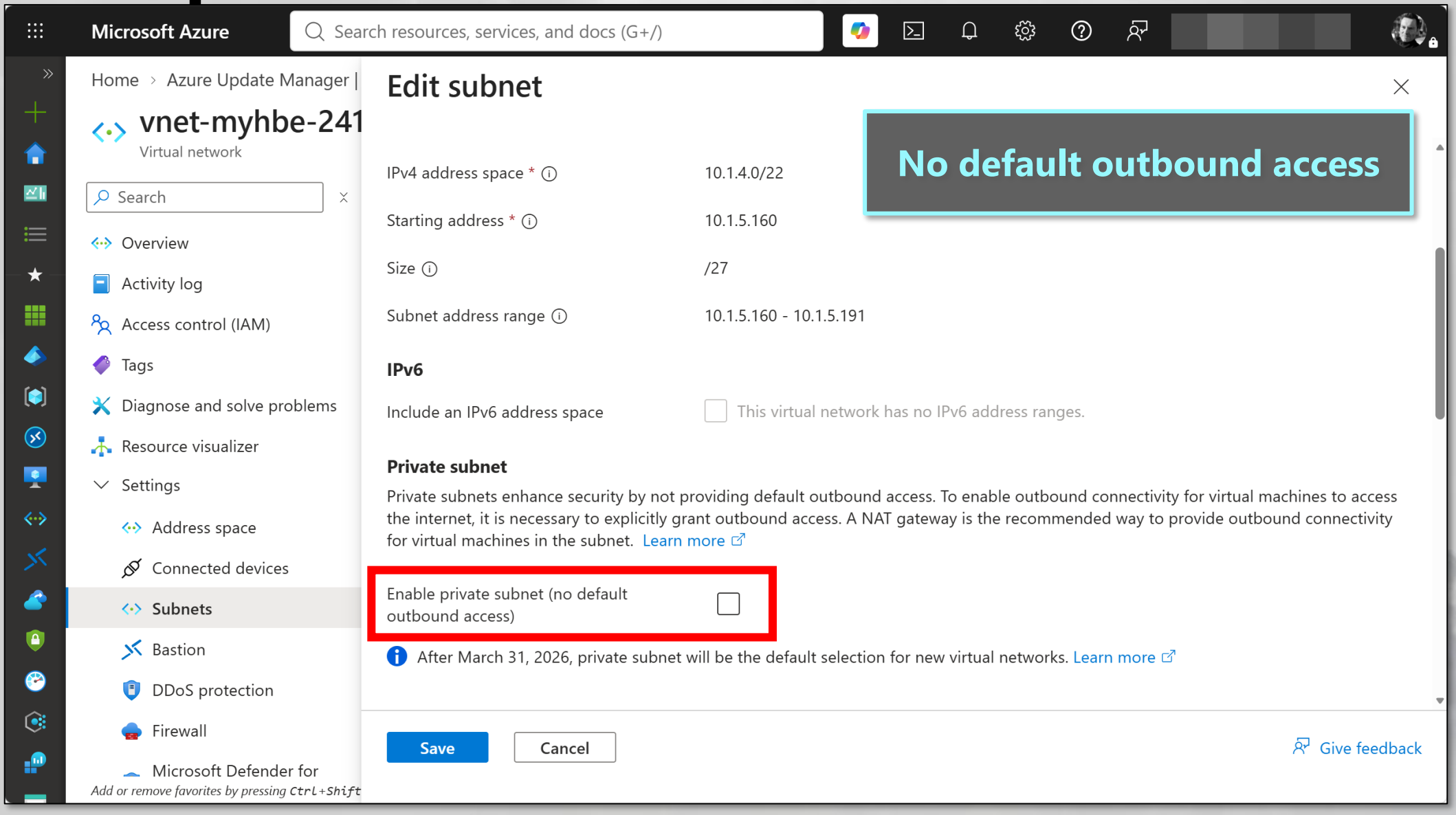


Azure NAT Gateway

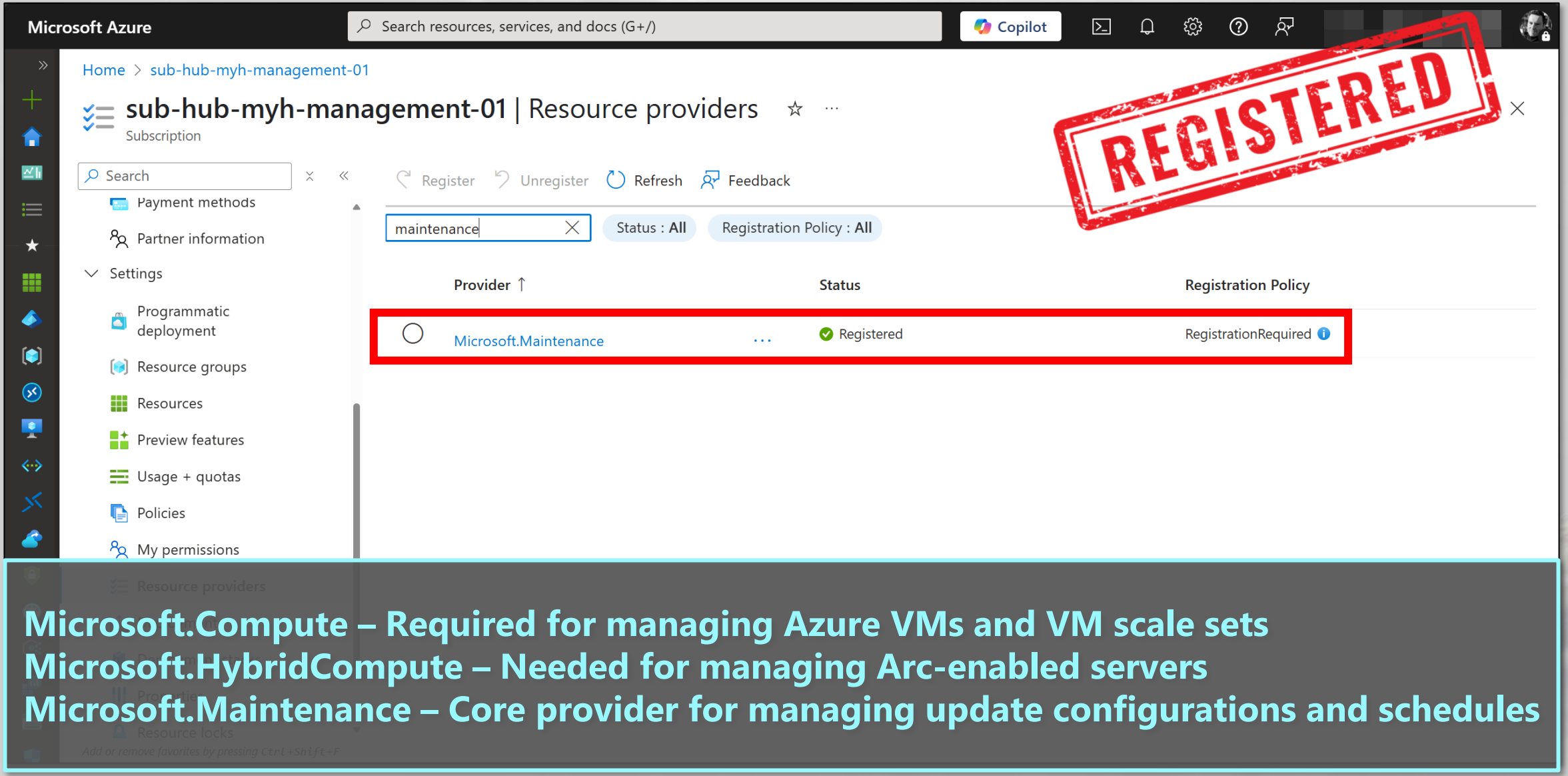
Your VM has a default outbound IP



Enable private subnet

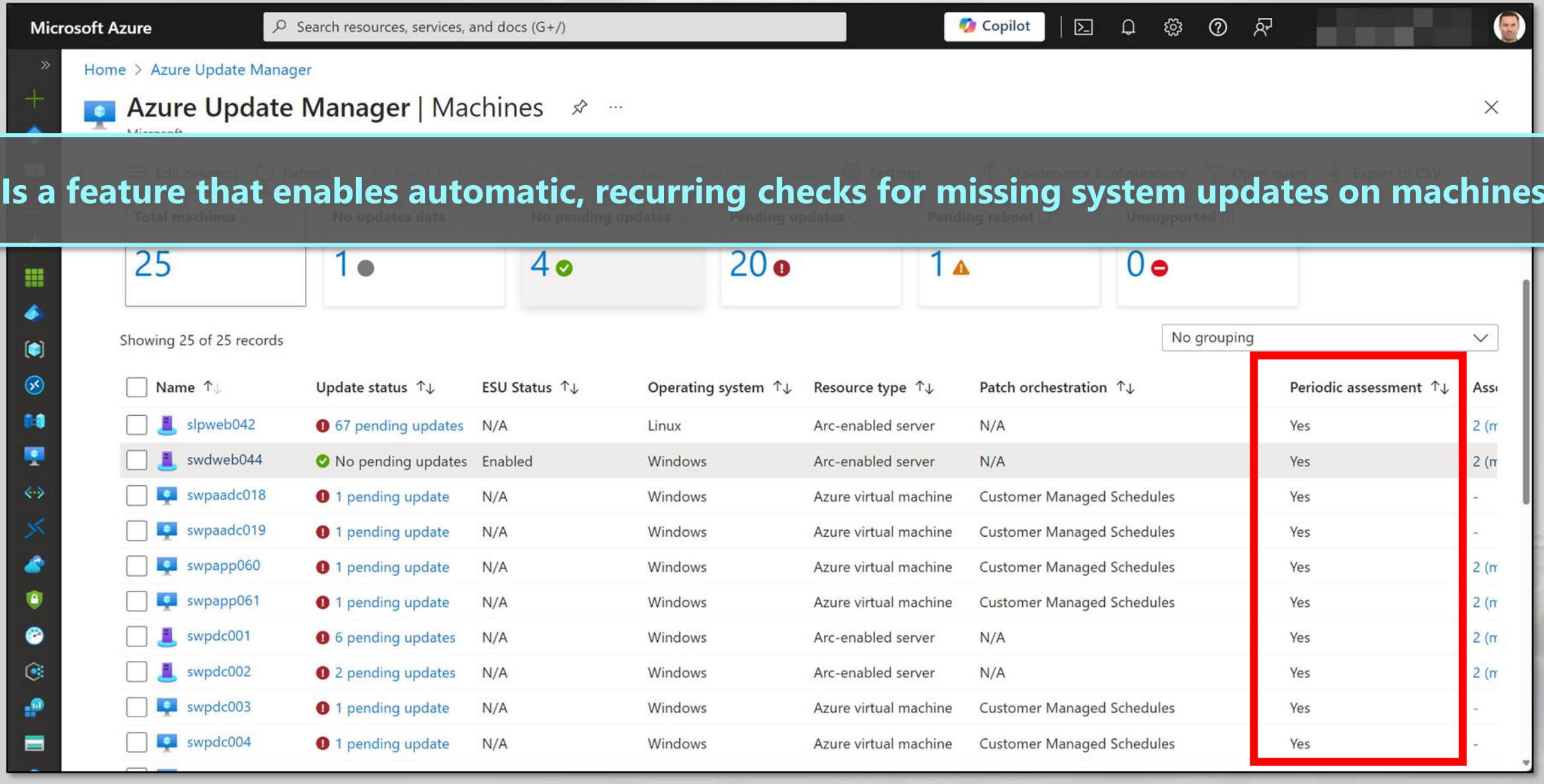


Prerequisites: Resource providers

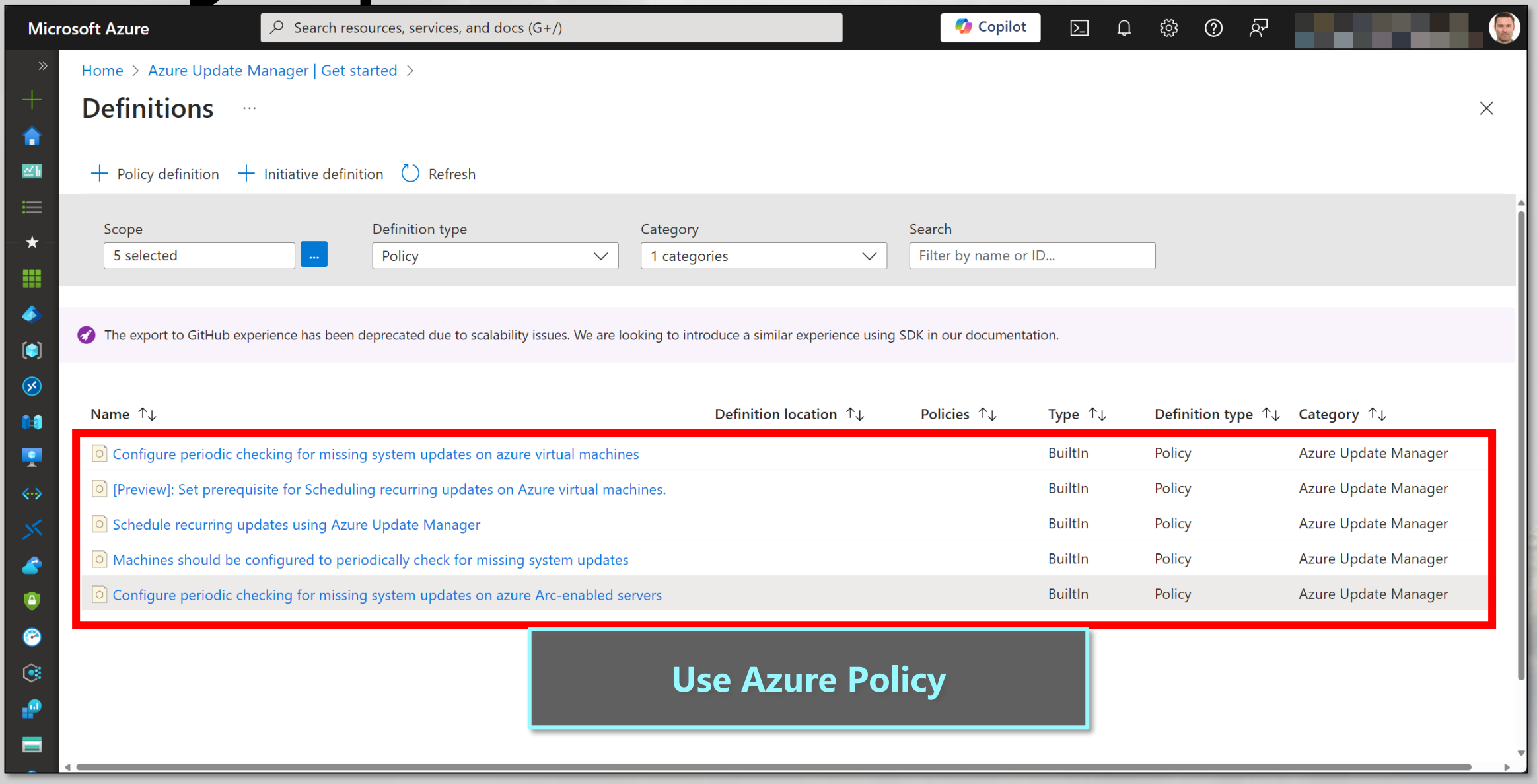


Microsoft.Compute – Required for managing Azure VMs and VM scale sets
Microsoft.HybridCompute – Needed for managing Arc-enabled servers
Microsoft.Maintenance – Core provider for managing update configurations and schedules

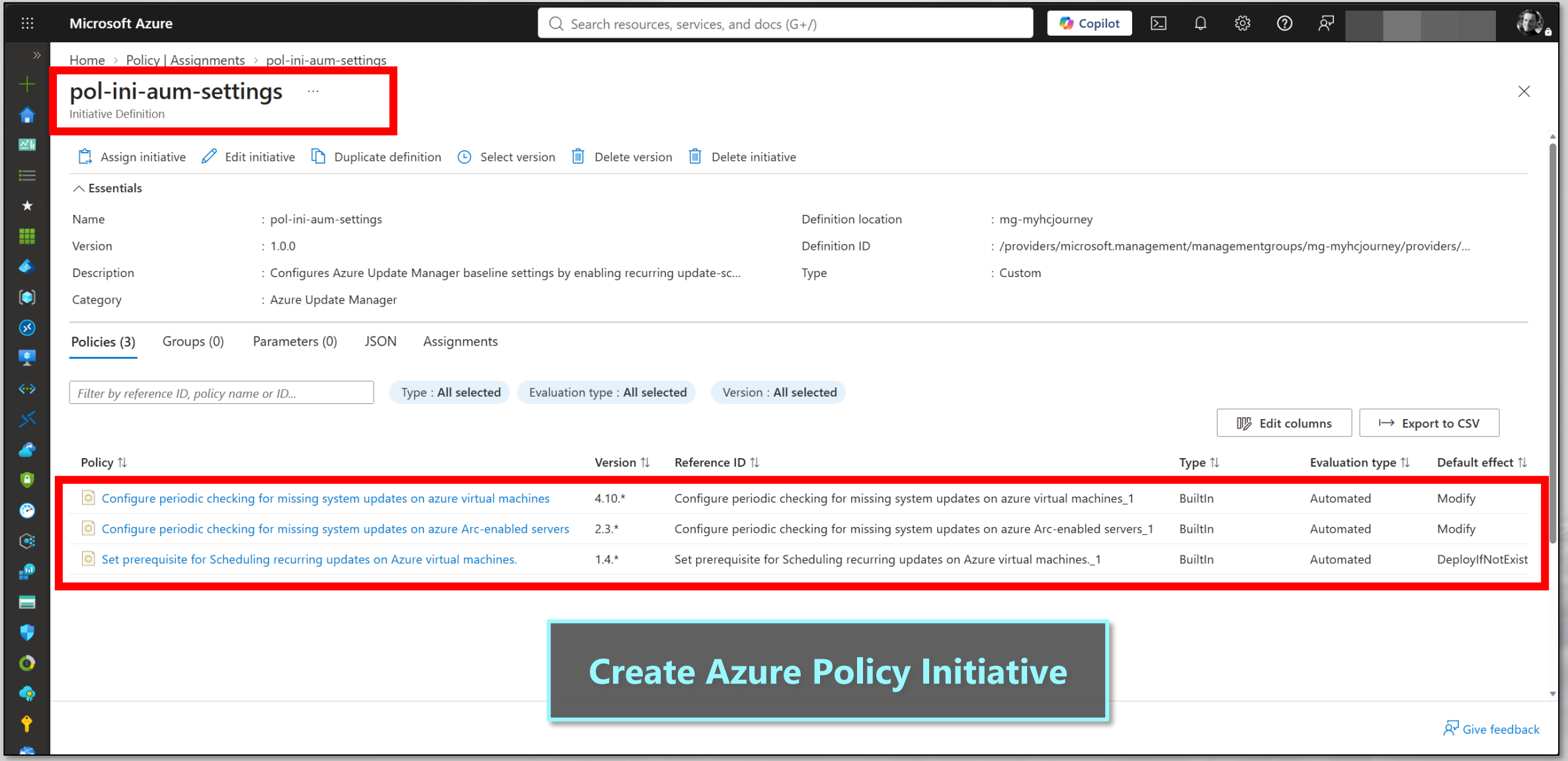
Periodic assessment



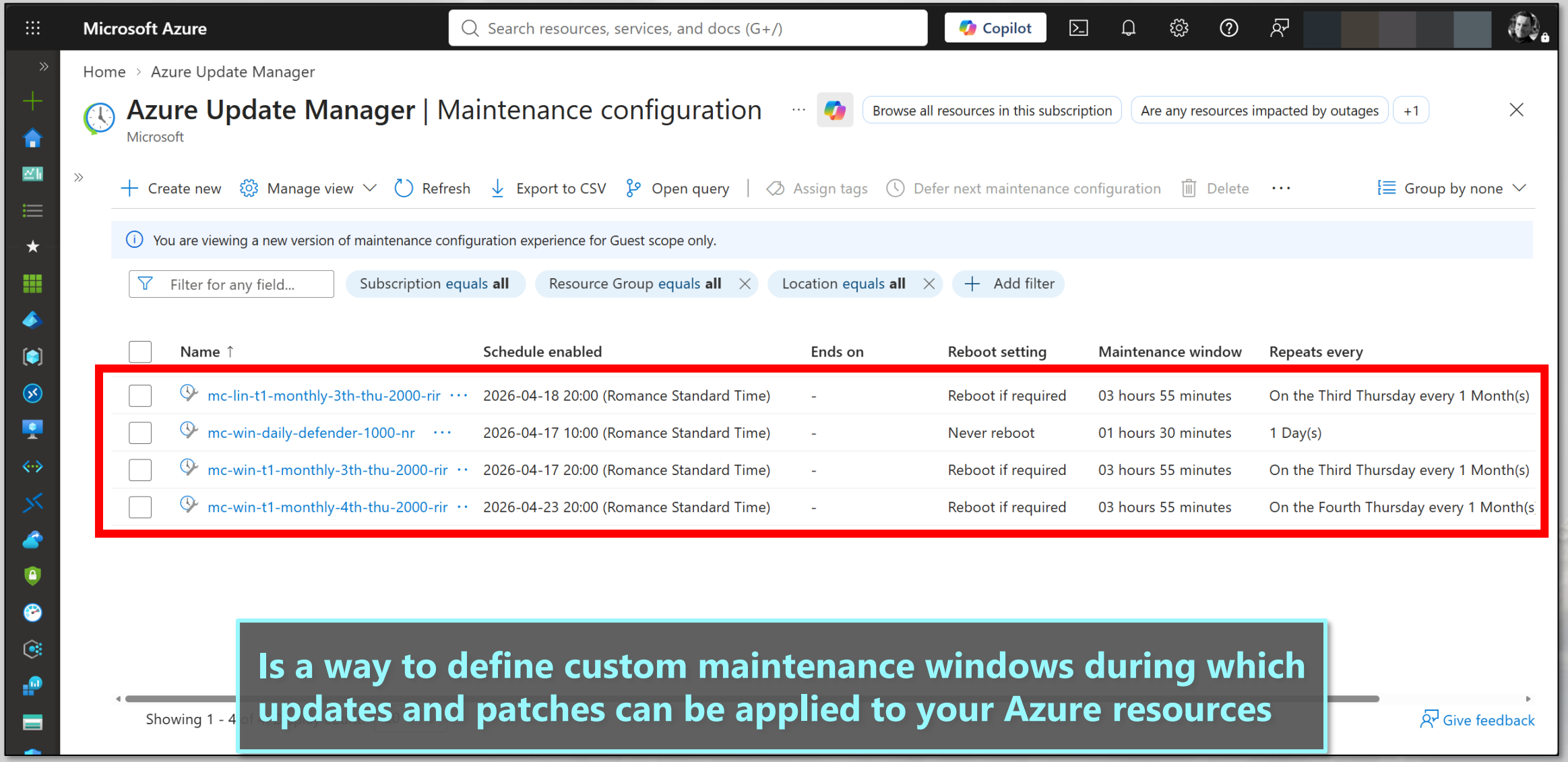
Configure periodic assessment



Configure periodic assessment



Maintenance Configurations



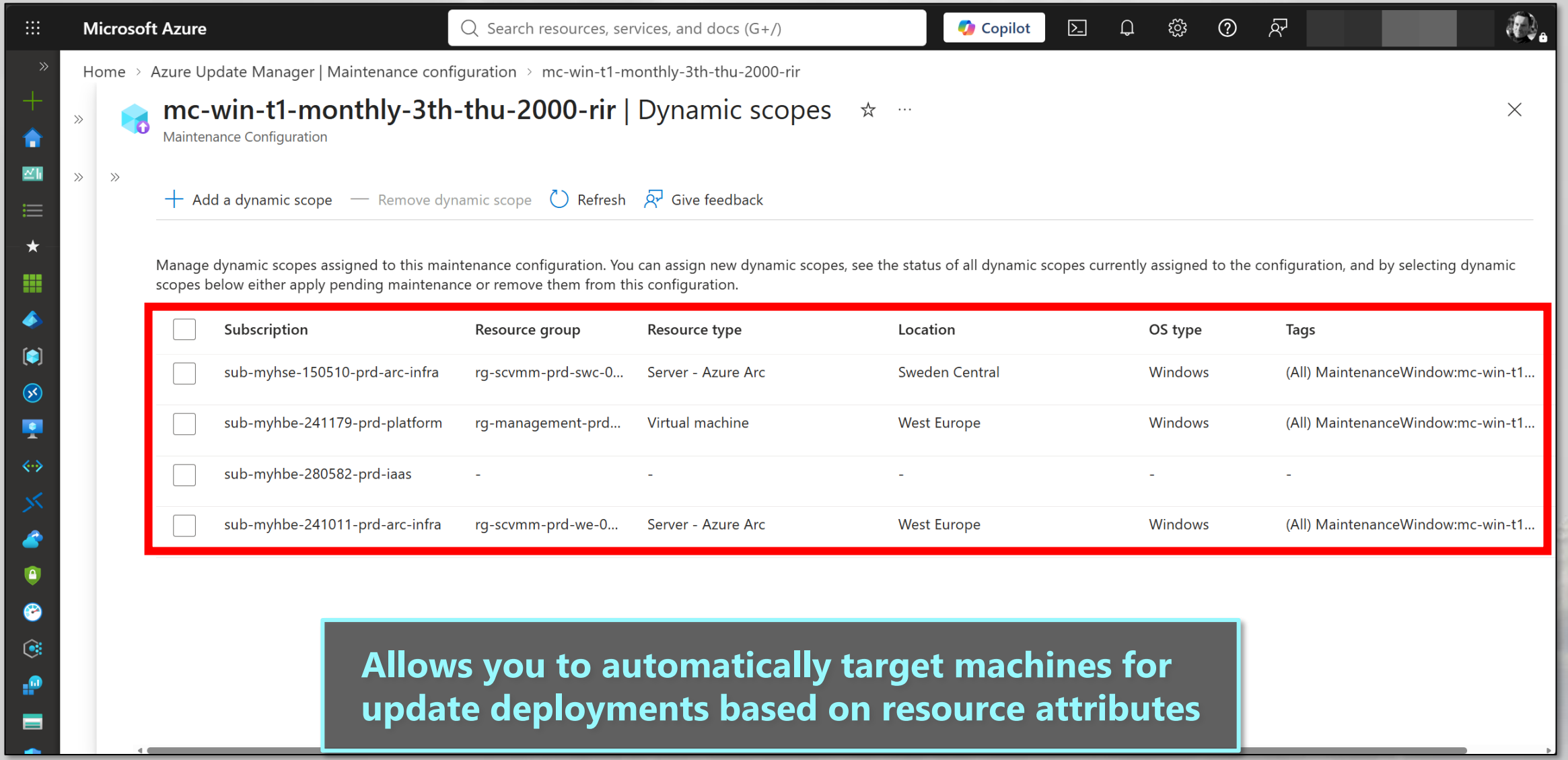
Is a way to define custom maintenance windows during which updates and patches can be applied to your Azure resources

Control what, who, when

- When updates can be applied (e.g., every Thursday at 10 PM)
- How long the maintenance window lasts (duration)
- Which resources are associated with the schedule (scope)
- Choose specific update classifications (e.g., security, critical, feature updates
- Select individual updates to include or exclude
- Schedule one-time or recurring update deployments



Dynamic scope



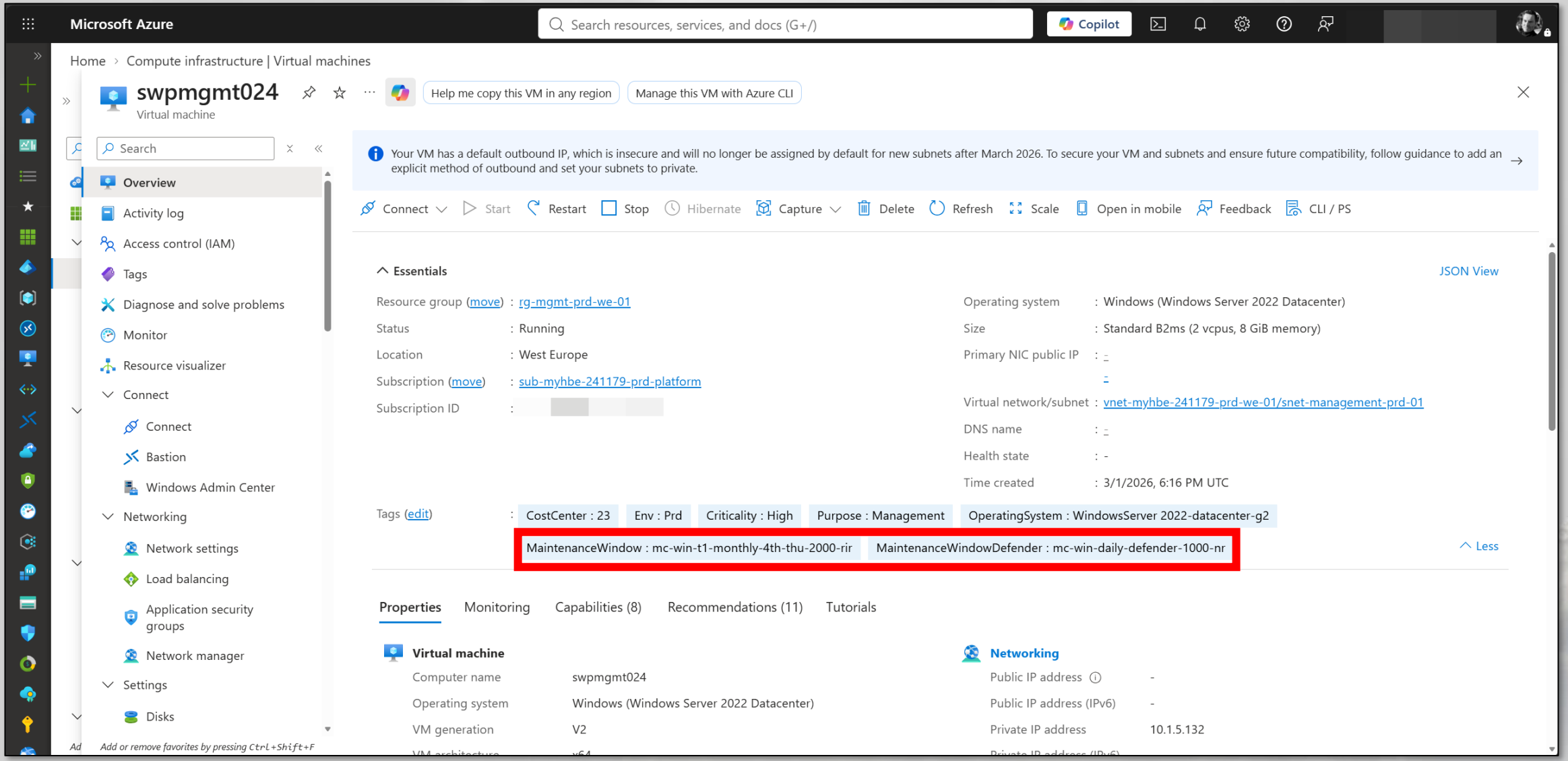
Allows you to automatically target machines for update deployments based on resource attributes

Dynamic scope

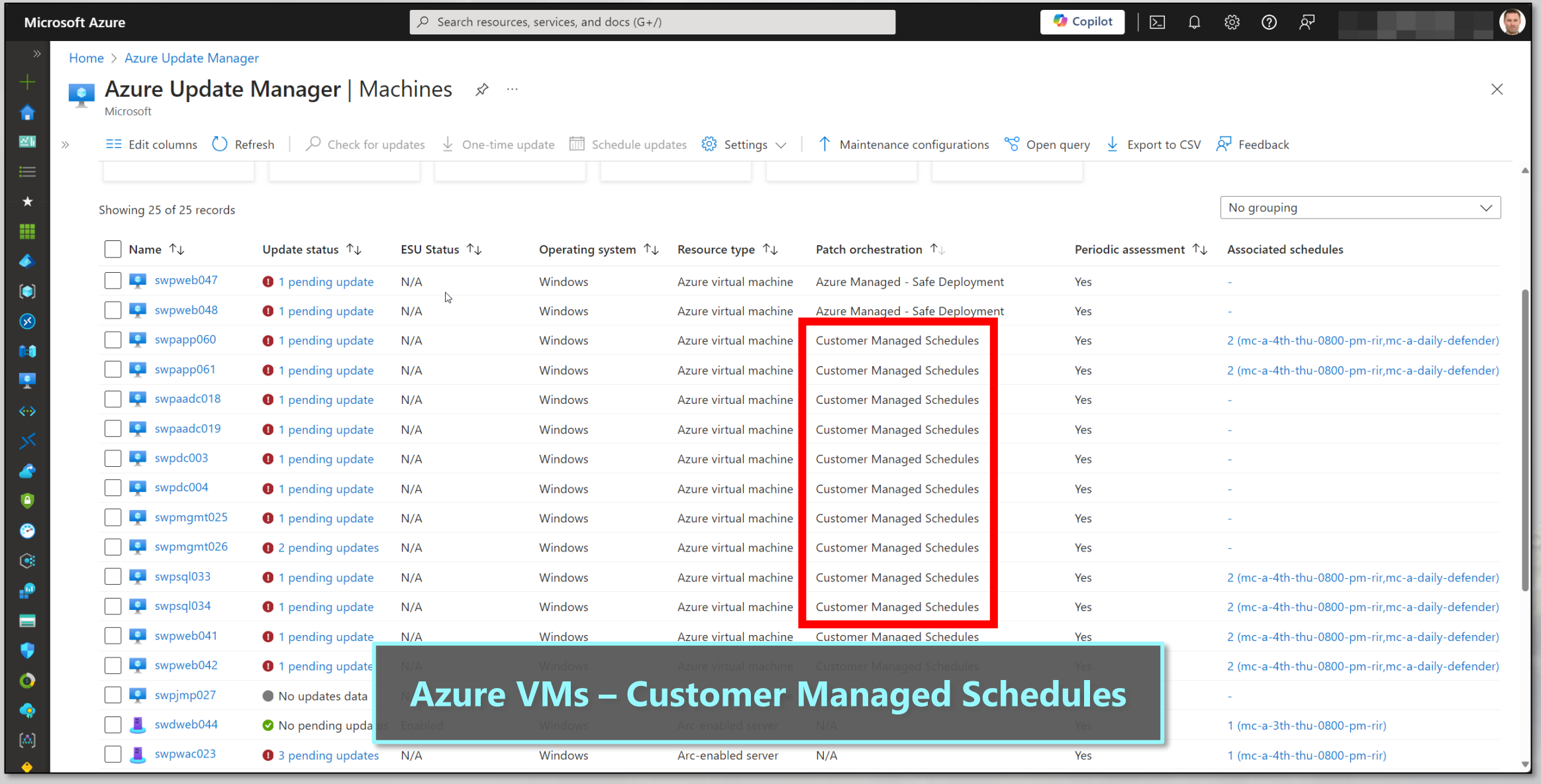
Uses Azure Resource Graph queries to define rules for selecting machines

- Tags (e.g. MaintenanceConfiguration : mc-win-t1-monthly-3th-thu-2000-rir)
- Resource group
- Resource types (Virtual machine, Server – Azure Arc)
- Location
- OS Type (Windows, Linux)

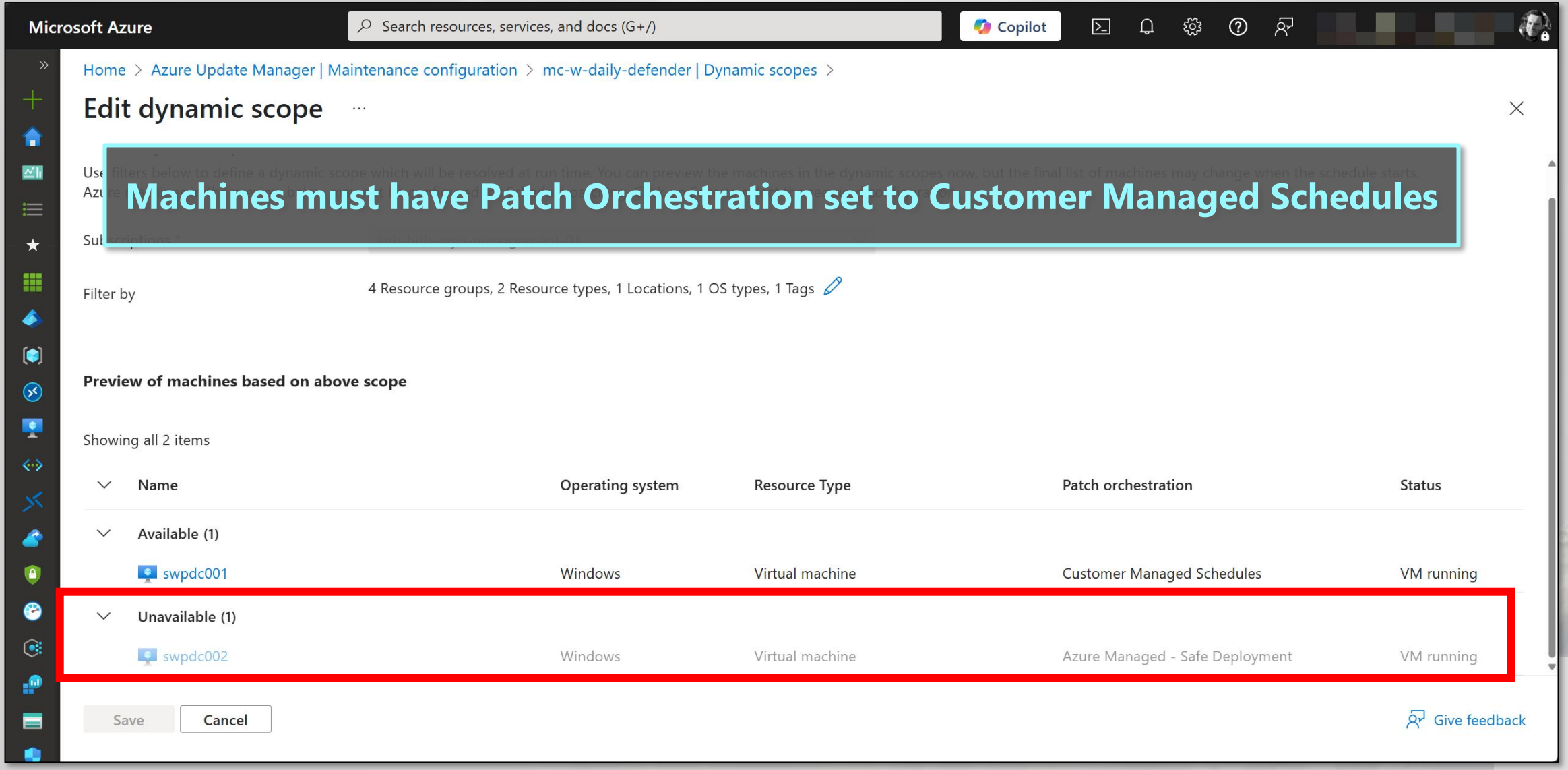
Dynamic scope based on Tags



Patch orchestration

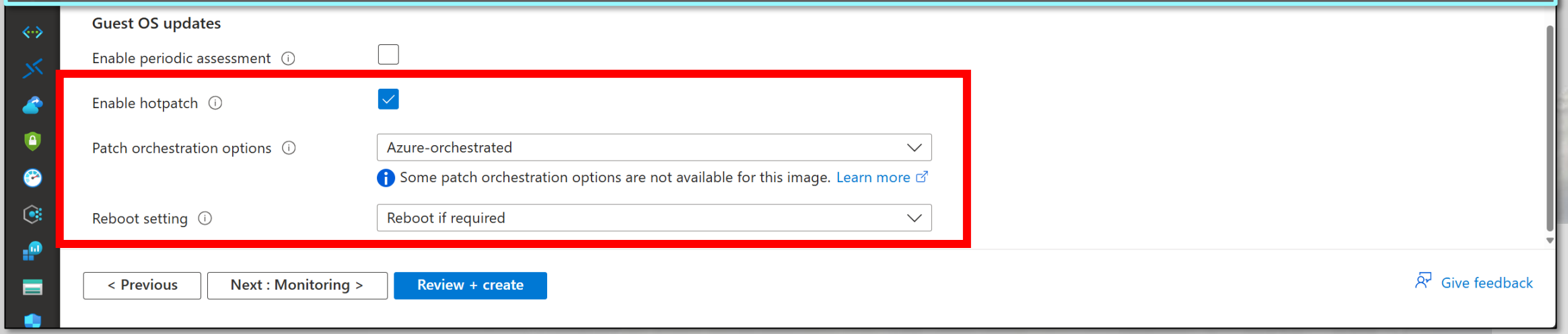


Enable Machine Preview in Dynamic scope



Enable Machine Preview in Dynamic scope

```
"osProfile": {
        "windowsConfiguration": {
                "patchSettings": {
                        "patchMode": "AutomaticByPlatform",
                        "assessmentMode": "AutomaticByPlatform",
                        "bypassPlatformSafetyChecksOnUserSchedule": true
                }
        }
}
```



Pricing 💵

- Free for Azure servers 🥳
- € 4,294 per server per month for Arc-enabled servers



Pricing 

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home > Microsoft Defender for Cloud | Environment settings >

Settings | Defender plans

sub-prd-myh-arc-infra-01

Search

Settings

- Defender plans
- Security policies
- Email notifications
- Workflow automation
- Continuous export

Save | Settings & monitoring

Cloud Workload Protection (CWP)

Microsoft Defender for Cloud provides comprehensive, cloud-native protections from development to runtime in multi-cloud environments.

Plan	Pricing*	Resource quantity	Monitoring coverage	Status
Servers	Plan 2 ($15/Server/Month) Change plan >	4 servers	Partial Settings >	Off / On
App Service	$15/Instance/Month Details >	0 instances		Off / On
Databases Action required	Selected: 1/4 Select types >	0 instances	Partial Settings >	Off / On
Storage	$10/Storage account/month $0.15/GB scanned for On-Upload Malwa Details >	0 storage accounts		Off / On
Containers	$6.8693/VM core/Month Details >	0 container registries; 0 kuberne	rtial gs >	Off / On
AI Services	$0.002/1K tokens/month Details >	0 AI resources		Off / On

Microsoft Defender for Cloud Servers Plan 2

Add or remove favorites by pressing Ctrl+Shift+F

Pricing 

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home > Azure Arc | Machines > swprds061

swprds061 | Windows Server

Machine - Azure Arc

Search

- Run command (preview)
- SQL Server Configuration
- Updates
- Inventory
- Change tracking
- Licenses
  - Windows Server
- Windows management
  - Windows Admin Center (preview)
  - Azure Site Recovery configuration (preview)
  - Best Practices Assessment (preview)
- Monitoring
  - Insights

Add or remove favorites by pressing Ctrl+Shift+F

License details

License type	Retail
License status	Licensed
Arc agent status	Connected

Azure benefits

If your machine license is covered by active Software Assurance, you can access key management services at no additional cost. Learn more

Activate Azure benefits: By checking this box, you attest that your Windows Server licenses have active Software Assurance or your Windows Server licenses are active subscription licenses.

Pay-as-you-go

Pay for your Windows Server license monthly on your Azure subscription. You can cancel at any time. Windows Server Pay-as-you-go includes the same Azure Benefits that can be activated by Windows Server Software Assurance customers. Learn more

Pay-as-you-go with Azure: I want to license Windows Server operating system on this machine using Pay-as-you-go.

Cannot activate Pay-as-you-go because machine must be unlicensed to be eligible.

Confirm | Cancel

Azure benefits

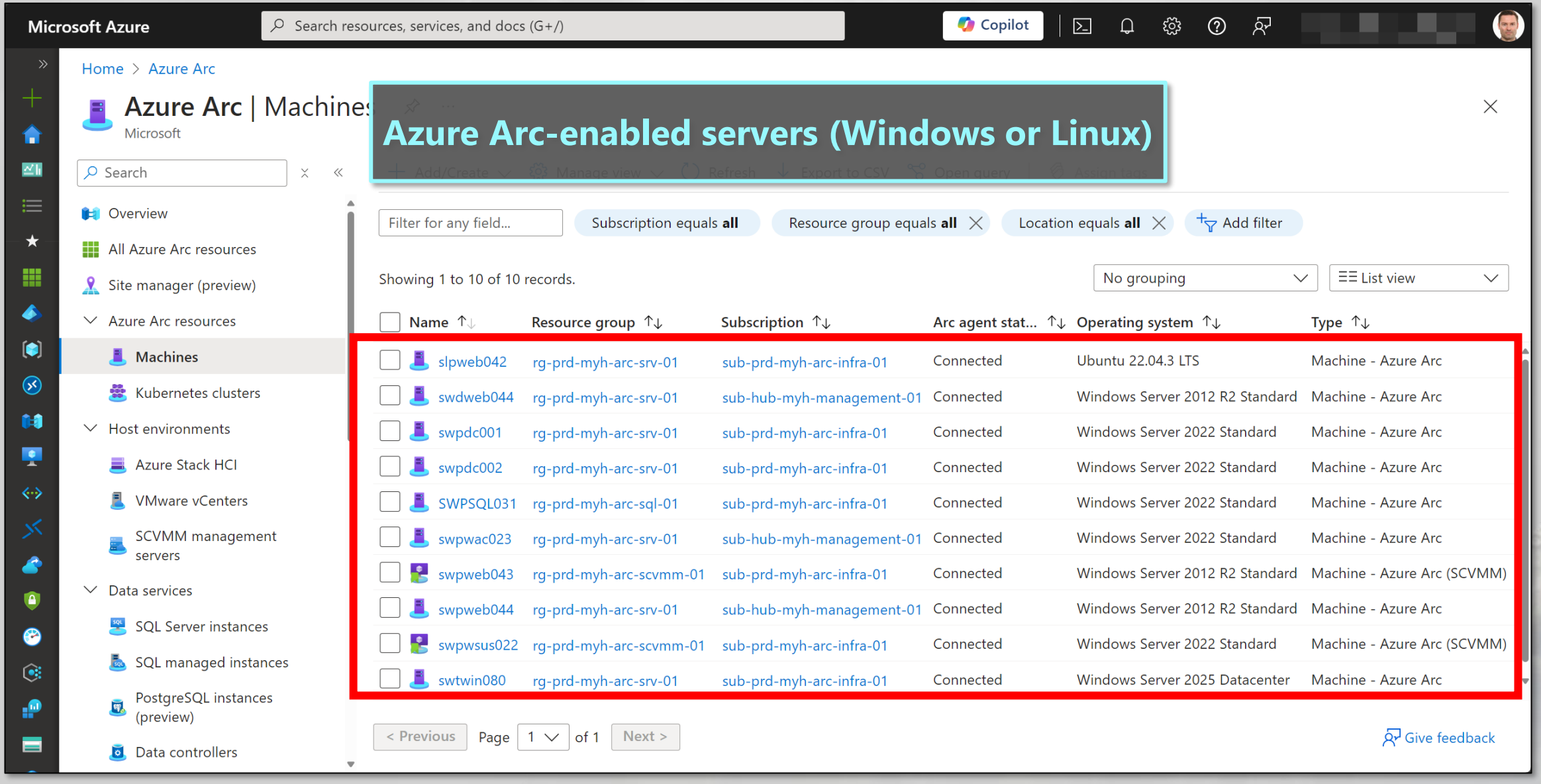
What are we still missing?

- Third-party or application-level patching
- No built-in rollback
- …

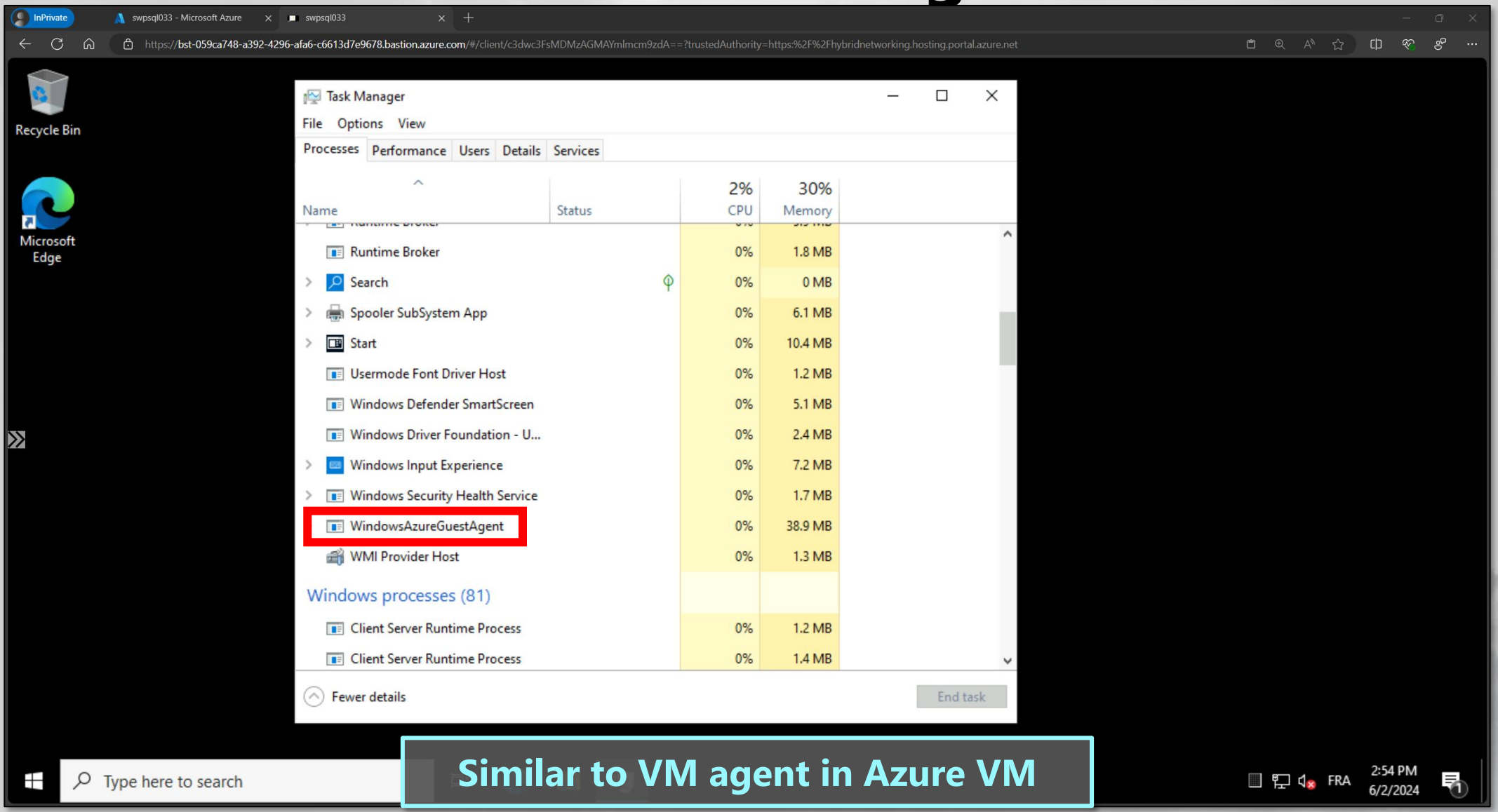


And what about on-premises?

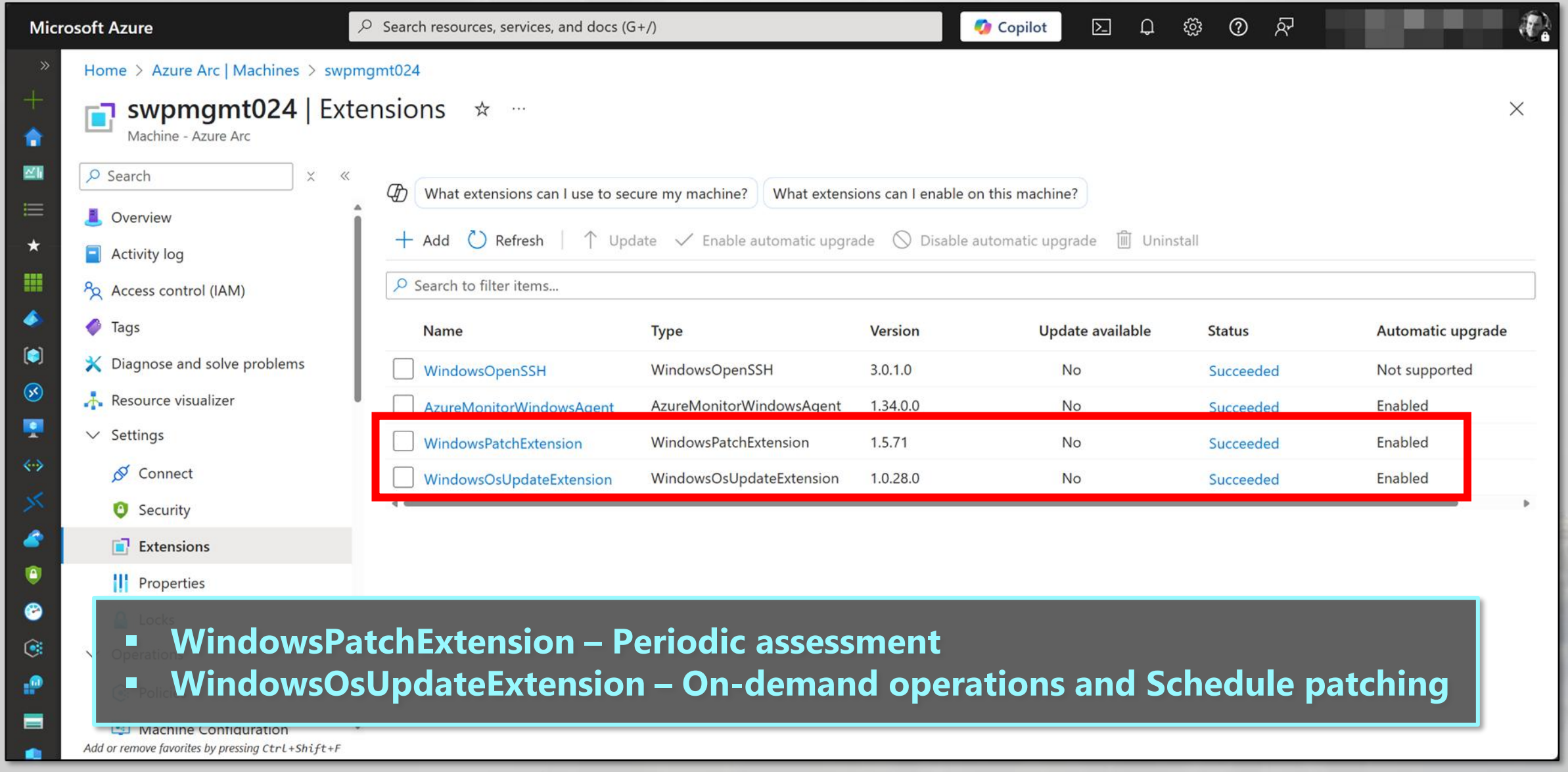
Azure Arc-enabled machine



Azure Connected Machine agent

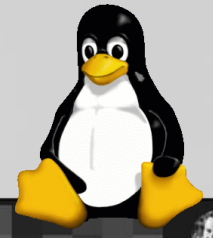


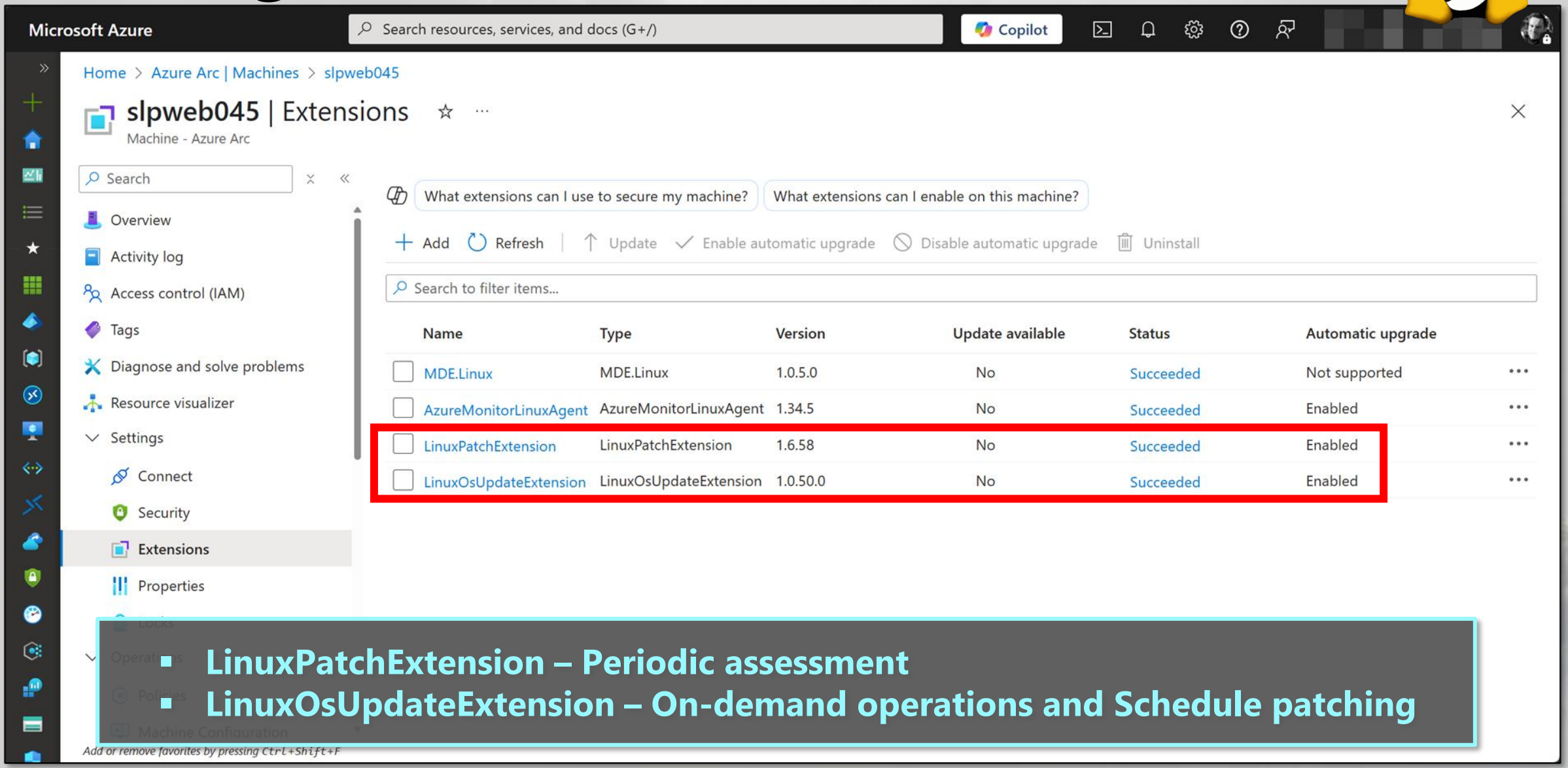
Azcmagent + extensions



- **WindowsPatchExtension – Periodic assessment**
- **WindowsOsUpdateExtension – On-demand operations and Schedule patching**

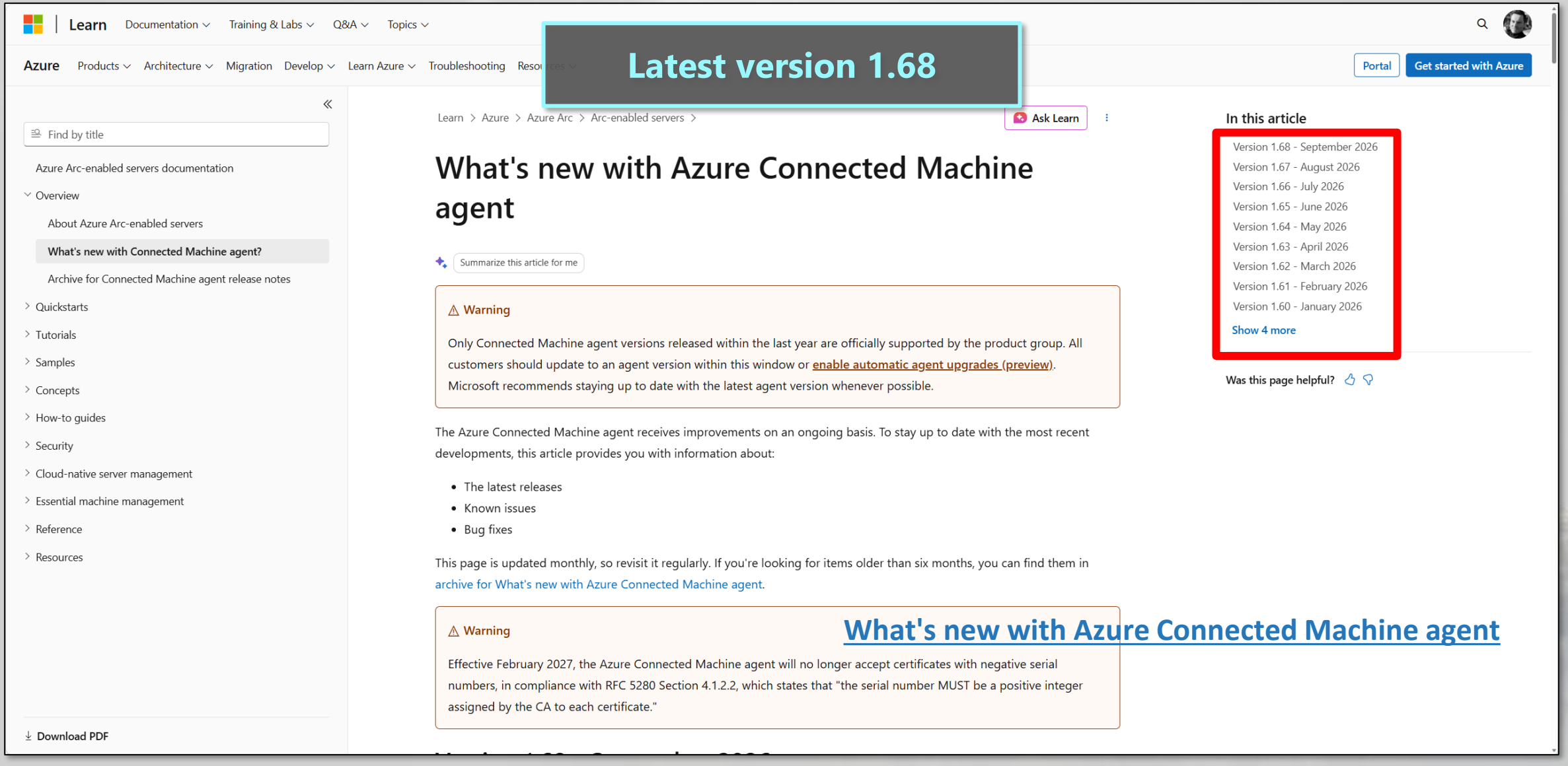
Azcmagent + extensions



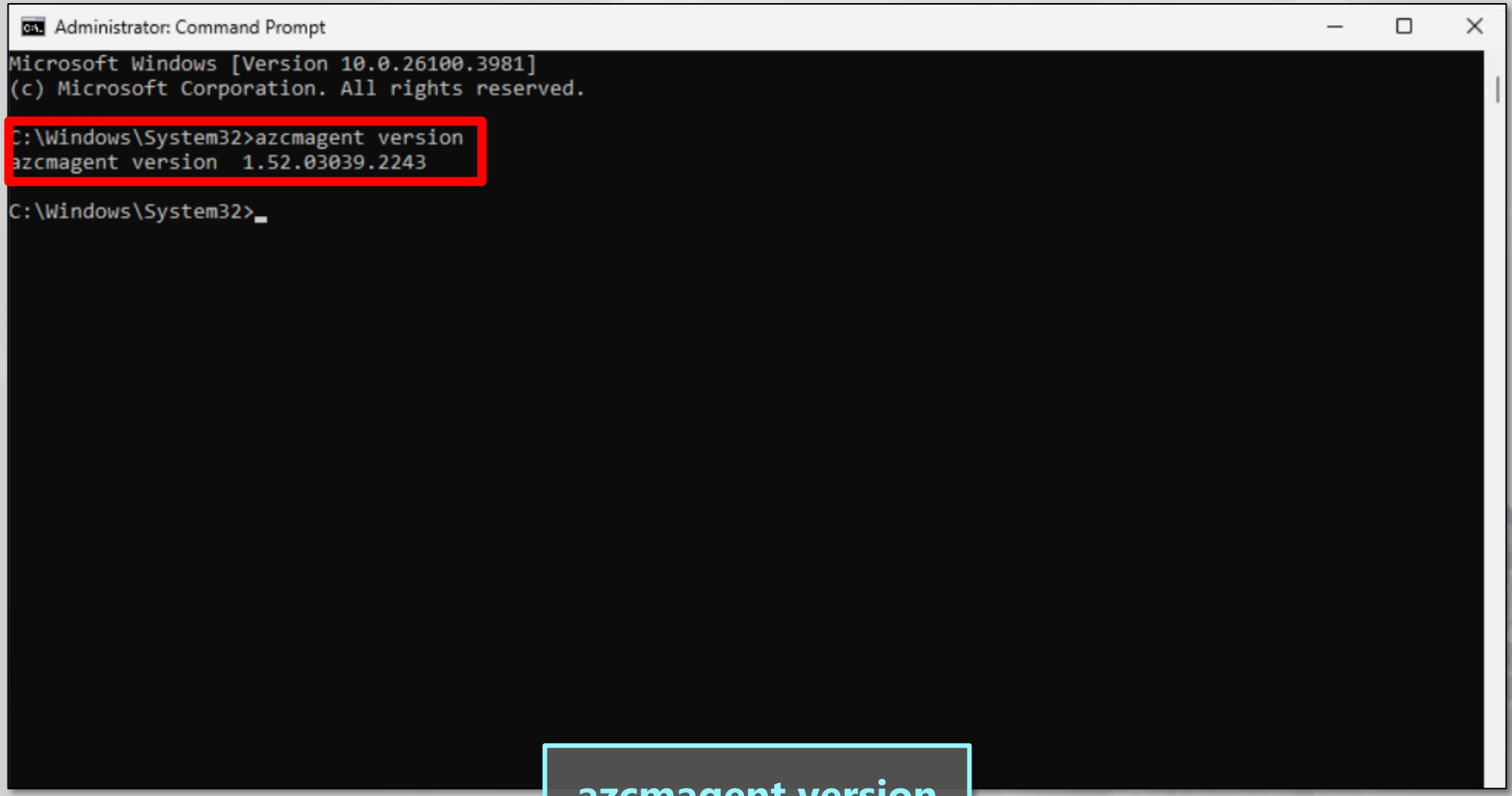


- **LinuxPatchExtension – Periodic assessment**
- **LinuxOsUpdateExtension – On-demand operations and Schedule patching**

Ensure your agent stays up to date

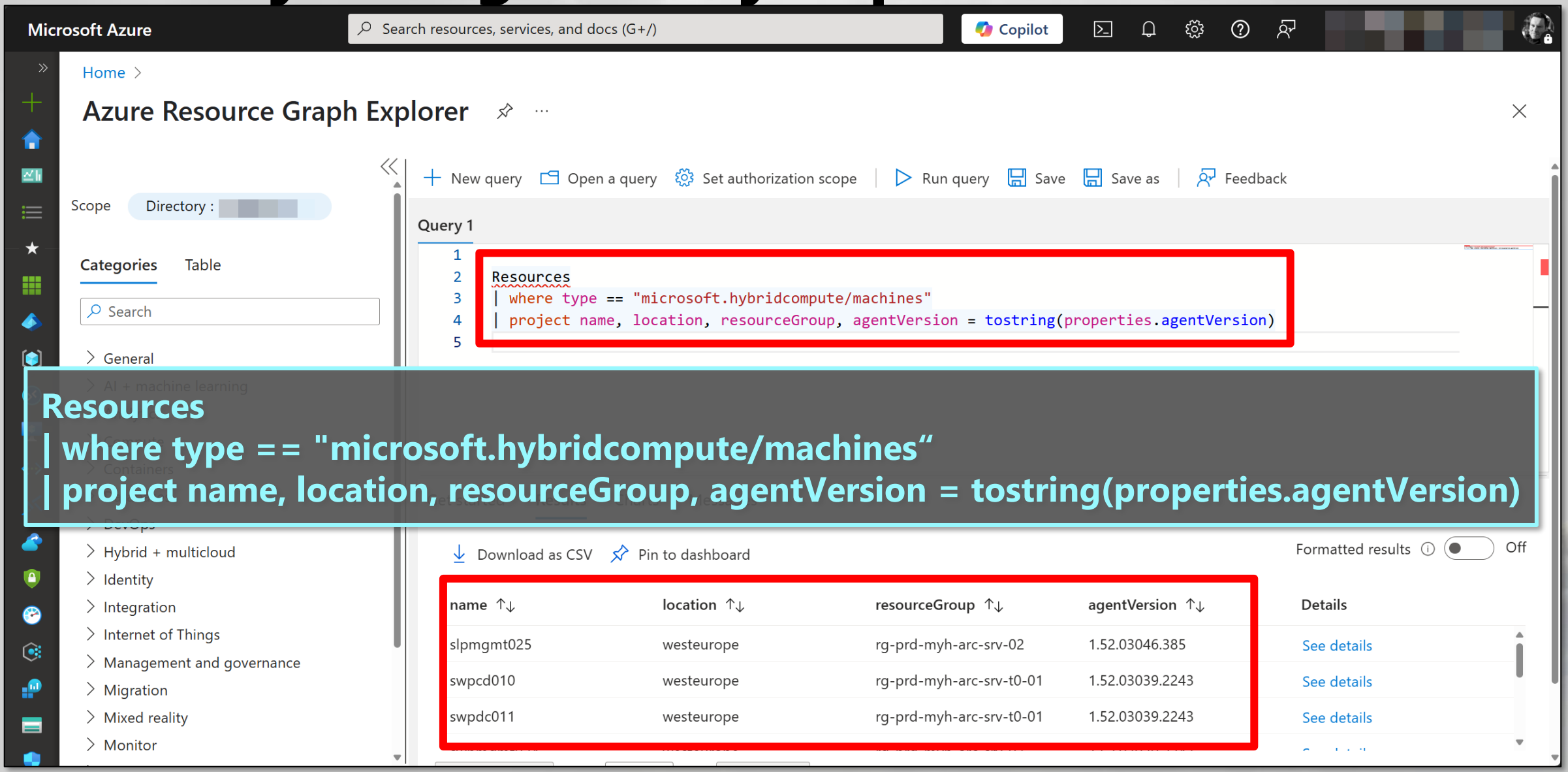


Ensure your agent stays up to date

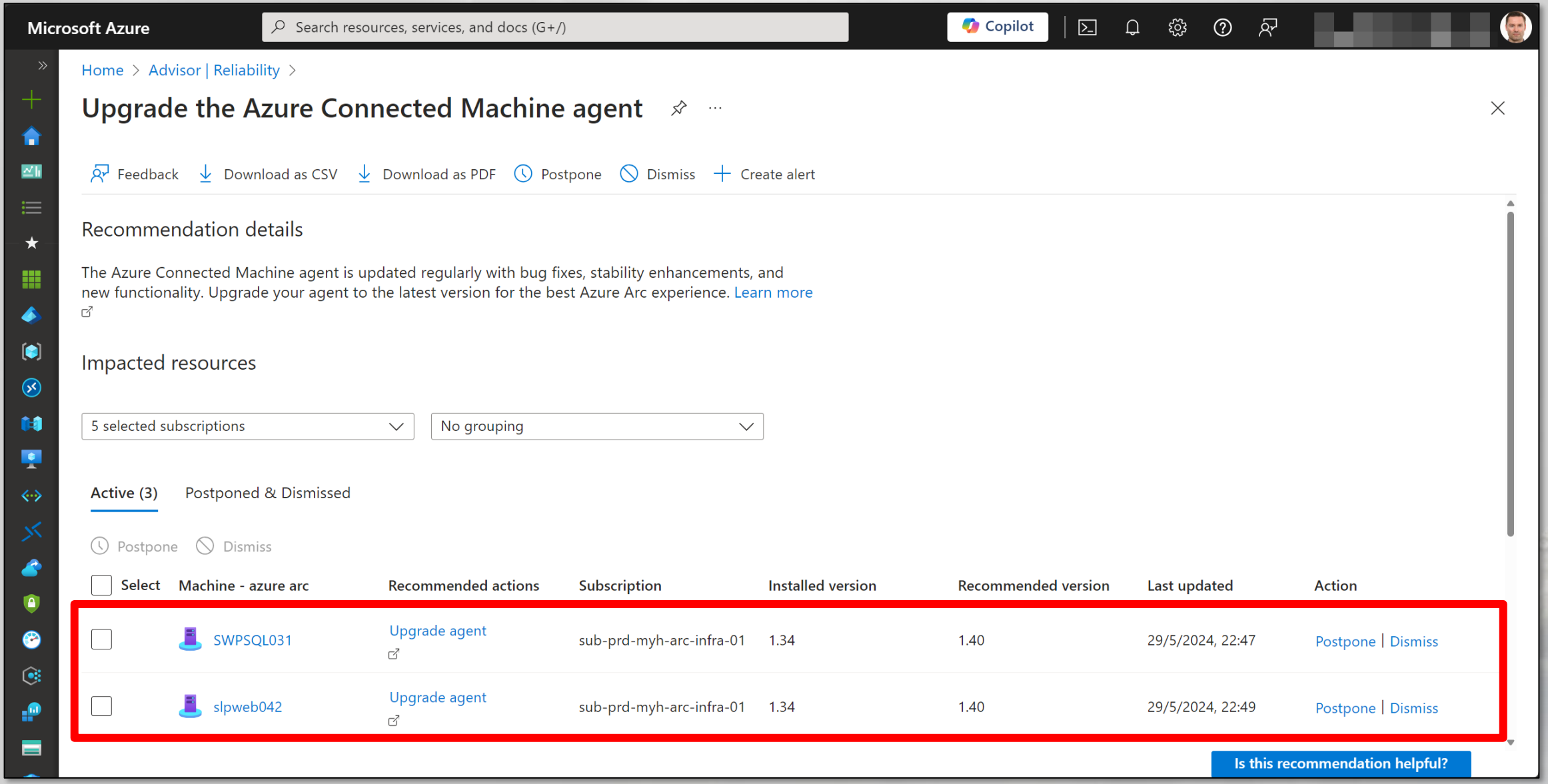


azcmagent version

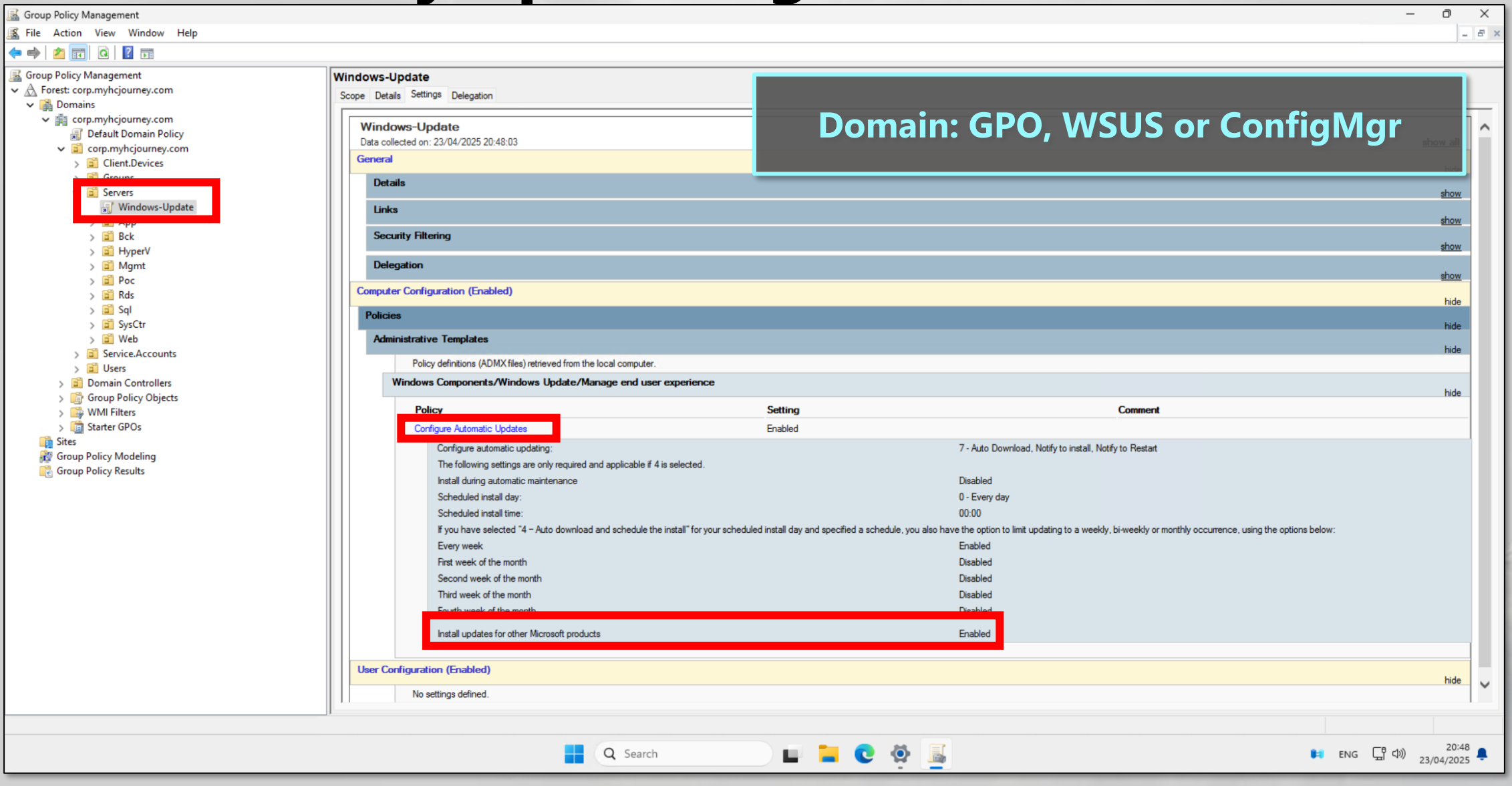
Ensure your agent stays up to date



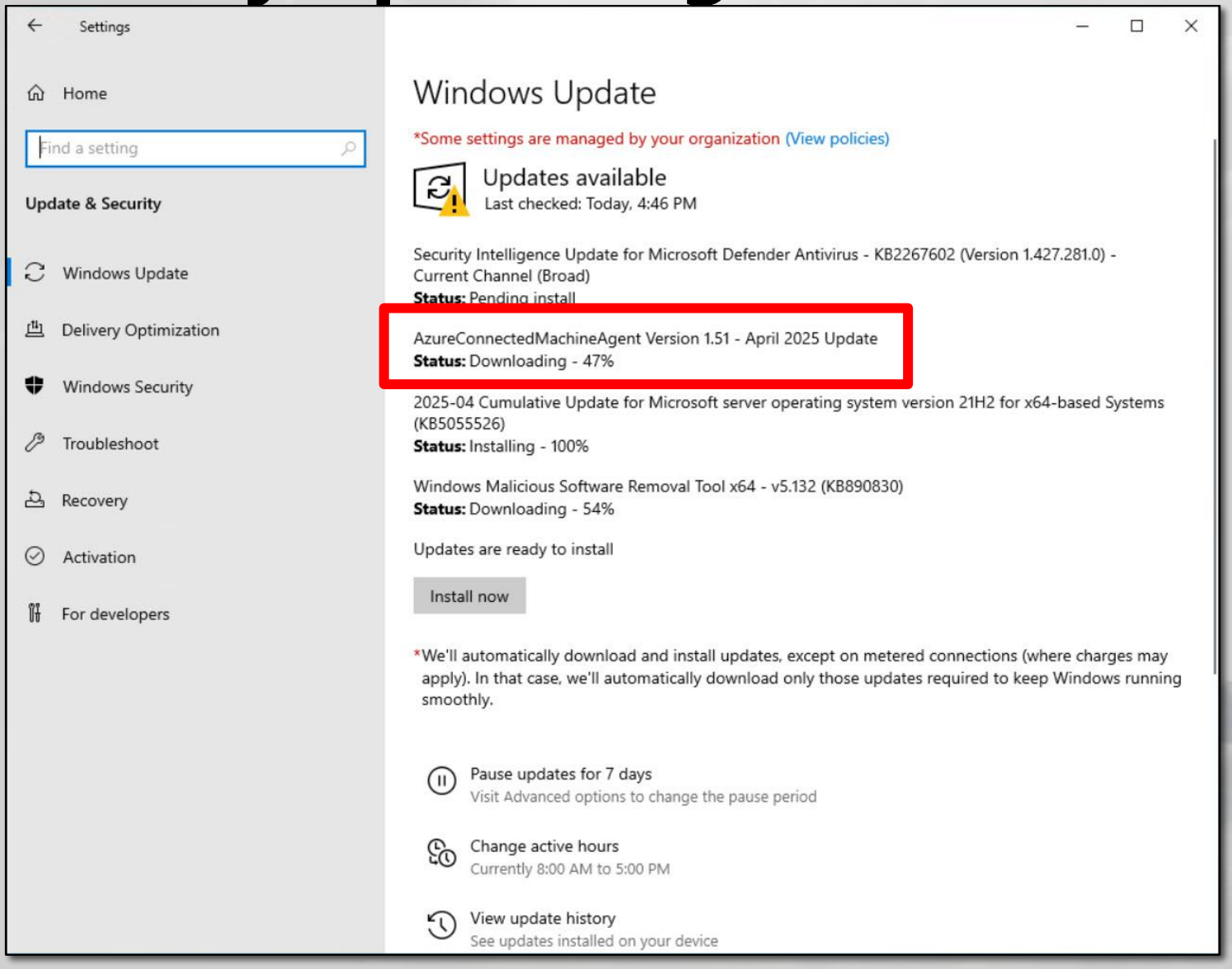
Azure Advisor



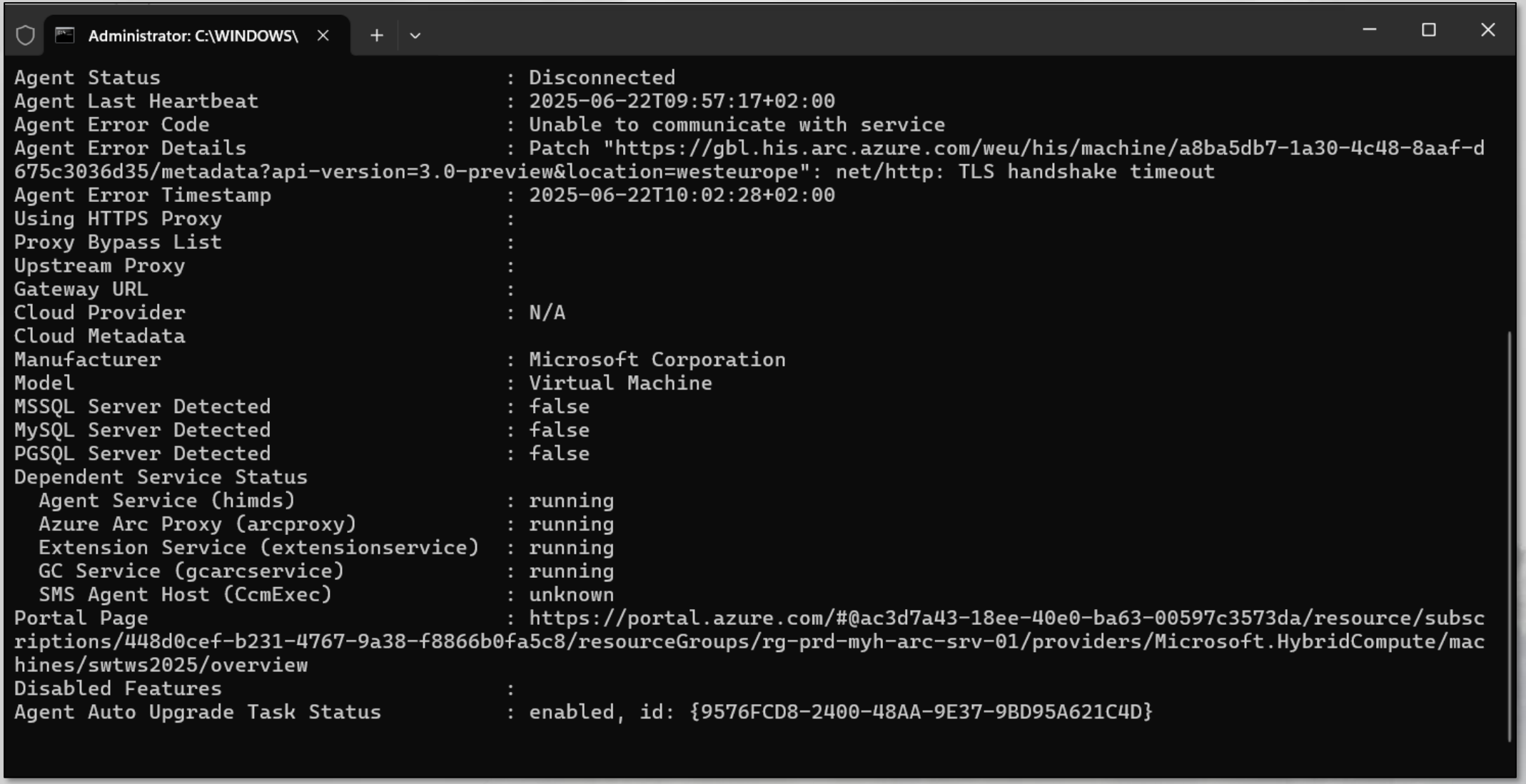
Automatically update agent



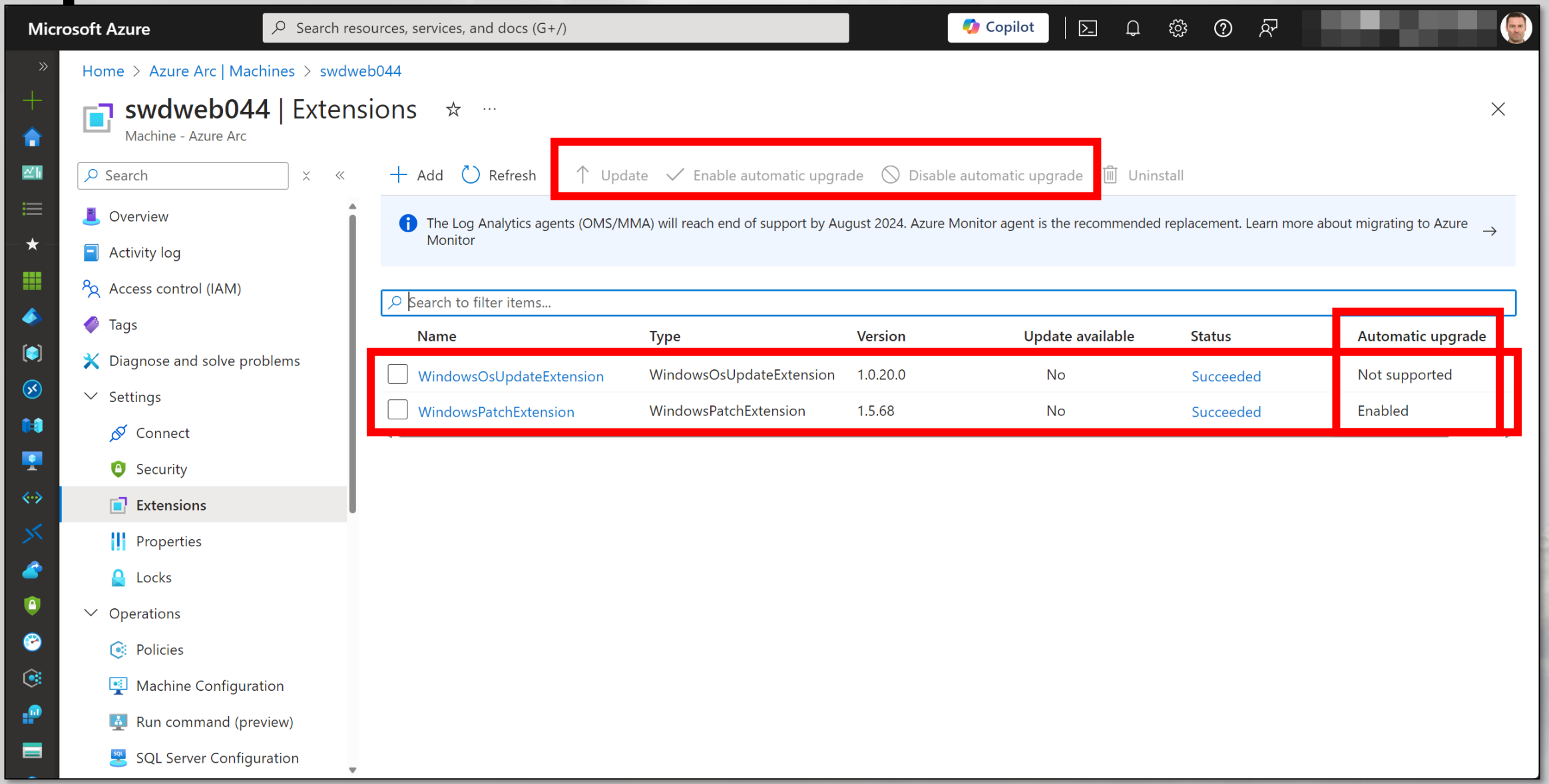
Automatically update agent



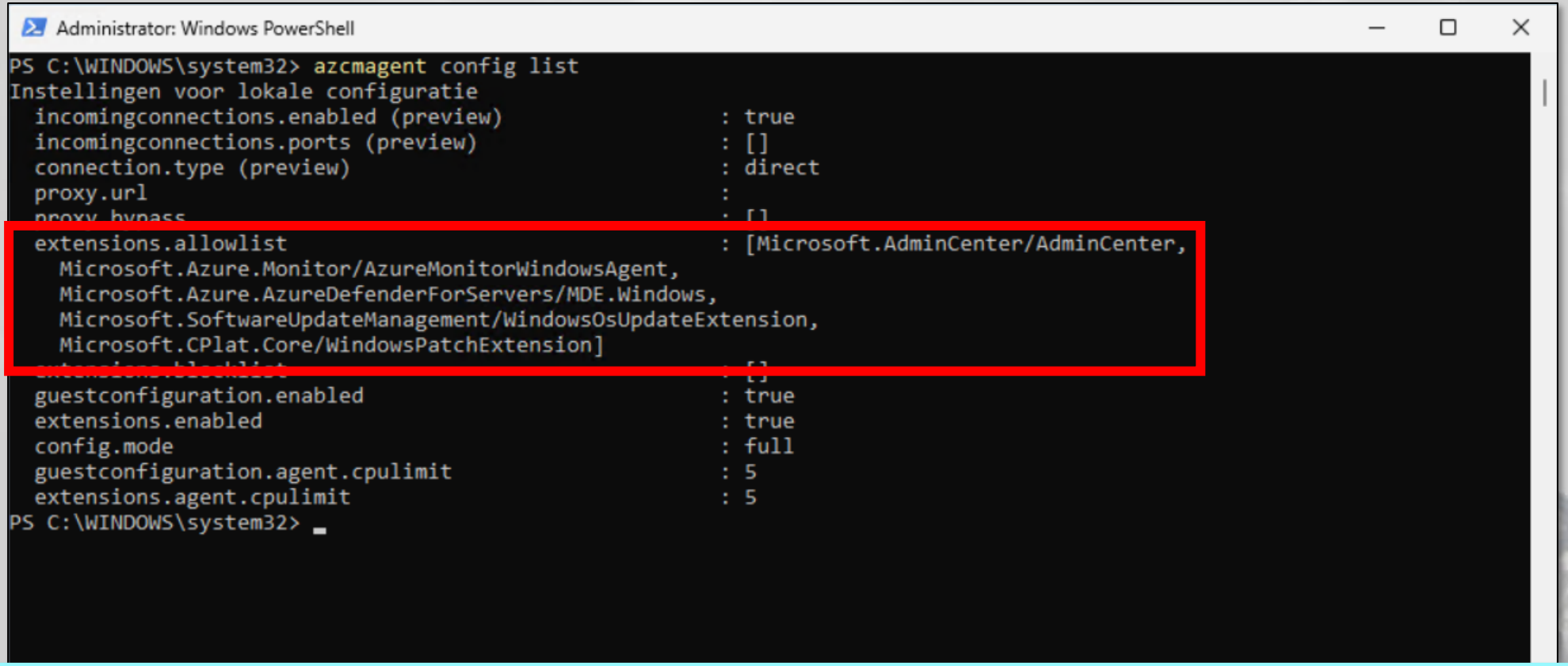
Agent Auto Upgrade



Update extensions

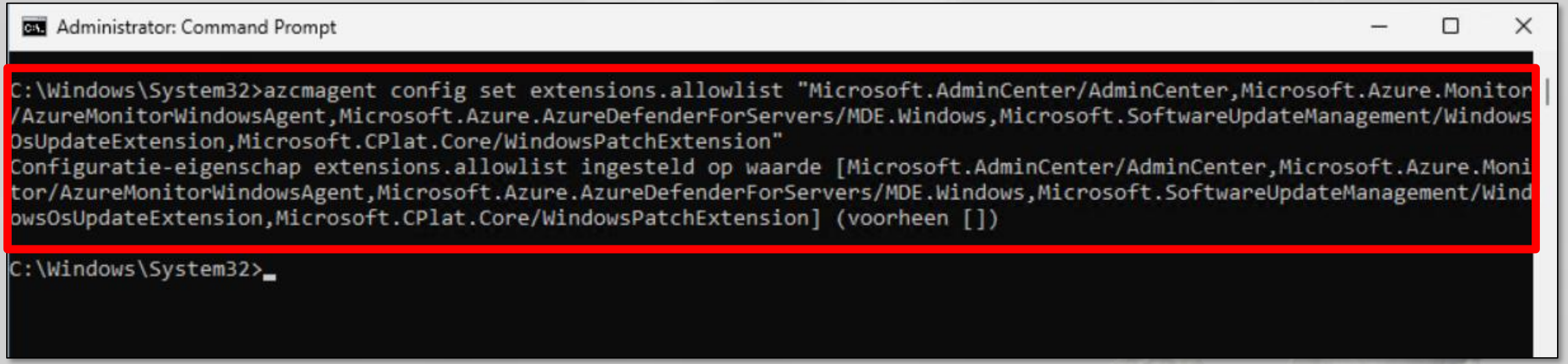


Extension allowlist for Azure Update Manager



- Periodic assessment - Microsoft.SoftwareUpdateManagement/WindowsOsUpdateExtension
- On-demand patching - Microsoft.CPlat.Core/WindowsPatchExtension

Adjust allowlist




```
azcmagent config set extensions.allowlist
"Microsoft.AdminCenter/AdminCenter,Microsoft.Azure.Monitor/AzureMonitorWindowsAgent,Mi
crosoft.Azure.AzureDefenderForServers/MDE.Windows,Microsoft.SoftwareUpdateManagement/
WindowsOsUpdateExtension,Microsoft.CPlat.Core/WindowsPatchExtension"
```



Thank you!

Wim Matthyssen

@wmatthyssen